Chapter I

Nonsmooth Optimization

1 Convex optimization

1.1 Directional Derivative

Remark. The function f does not to be continuous.

We need to establish a Existenzaussage about the directional derivative.

This is a very important auxiliary function because in the convex function the distance between any two points is upper bounded by this function.

The existence of directional derivative, the local Lipschitz continuity of convex function, the convexity of sub-differential, the equivalence of sub-differential and gradient for differentiable functions, and the equivalence of Clark derivative and directional derivative have all used this operator.

The sublinear function defined a half norm on the open set $D \subset \mathbb{R}^n$. Can we find a directional derivative, which for some $s,t \in \mathbb{R}^n$ admits strict sub-linearity? We now look into the sufficient and necessary conditions of minimal points $\hat{x}$.

Theorem 1.1 (NOB)**.** *Given a function: $f : D \longrightarrow \mathbb{R}$, where D is open. For the minimal point $\hat{x} \in D$, we have:*

1. *the necessary conditions: f admits directional derivative in every direction and it holds*

$$f'(\hat{x}, x-\hat{x}) \geq 0, \quad \forall x \in \mathbb{R}^n. \tag{I.1}$$

2. *The sufficient condition: f is convex and satisfies the condition above.*

Definition 1.1 (Star-shaped area)**.**

Definition 1.2 (Lipschitz continuity)**.**

Remark. Convexity implies locally Lipschitz-continuity.

Theorem 1.2. *Given a convex $f : D \longrightarrow \mathbb{R}^n$, D is open and convex.Then we have:*

1. *$\forall x \in D$, the subdifferential $\partial f(x)$ is non-empty, convex, and compact.*

2. *If f is Lipschitz continuous, we have: $\partial f(x) \subset U_L(x)$, and*

$$f'(x,s) = \max_{g \in \partial f(x)} g^\top s, \forall s \in \mathbb{R}^n \tag{I.2}$$

Proof. The difficult part is to prove that the subdifferential is not empty for the convex function and equation (6) using the separation theorem.

1. $\partial f(x)$ is closed and convex:
   By the definition of sub-differential we know that $M_y := \{g | f(y) - f(x) \geq g^\top (y-x)\}$ is convex for all y. Because the intersection of convex sets is also convex, we know the $\partial f(x)$ is convex.

2. the equation is shown by separation theorem.

□

Theorem 1.3. ***Given a convex function, a point is a minimum point iff 0 is included in the subdifferential:*** *For a convex function defined on a non-empty, open and convex set, the following arguments are equivalent:*

1. *The point $\hat{x}$ is local, and therefore global minimal point on the set D.*

2. $0 \in \partial f(\hat{x}))$

Here we have a generalisation of the necessary conditions of optimal points in the case of smooth function to the case of non-smooth(convex) functions.

Lemma 1.4. *The union of convex and compact sets is compact and we have:*

$$C = \{\sum_{i=1}^{m} \alpha_i x_i \mid x_i \in C_i, \alpha_i \geq 0, \sum_{i=1}^{m} \alpha_i = 1\}$$

Theorem 1.5. *Given a family of convex continuous function $f_i : \mathbb{R}^n \longrightarrow \mathbb{R}$. We define the convex function:*

$$f(x) := \max_{1 \leq i \leq m} f_i(x). \tag{I.3}$$

$\forall x \in \mathbb{R}^n$, define $I(x)$ the set of indices, of which the function is active in the point x. Than we have :

$$f'(x,d) = \max_{i \in I(x)} f_i'(x) \quad \forall d \in \mathbb{R}^n$$

And the subdifferential is the convex hull of the subdifferential active functions. i.e.

$$\partial f(x) = \text{conv}\{i \in \cup_{i \in I(x)} \partial f_i(x)\}$$

Proof. The existence of the directional derivative is guaranteed by the convexity of the function. By definition we get the form of (7)
To show $\partial f(x) = \text{conv}_{i \in I}\{\partial f_i(x)\}$
"$\subset$",
by definition we know:

$$g \top s \leq f_i'(x,s) \leq \partial f(x), \forall s$$
$$\Rightarrow \cup \partial f_i \subset \partial f$$
$$\overset{(2)}{\Rightarrow} \text{conv}\{\cup_{i \in I} \partial f_i\} \subset \partial f$$

(2) because the sub-differential is a convex subset of $\mathbb{R}^n$.
"$\supset$"

□

Corollary 1.5.1. *If the function $f_i : \mathbb{R}^n \longrightarrow \mathbb{R}, 1 \leq i \leq m$ are affine. i.e.*

$$f_i = a_i^T x + b_i$$

Define:

$$f = \max_{1 \leq i \leq m} f_i(x)$$

Then $\forall x, d \in \mathbb{R}^n$

$$f'(x,d) = a_i^\top d, \quad \partial f(x) = conv\{a_i | i \in I(x)\}$$

1.2 Derivative by Clark

Definition 1.3 (Clark derivative)**.** Given a function $f : D \longrightarrow \mathbb{R}^n, D \subset \mathbb{R}^n$ open. For $\hat{x} \in D, s \in \mathbb{R}$. The Clark derivative exists if:

$$f^\circ(\hat{x}, s) := \limsup_{t \to 0_+, x \to \hat{x}} \frac{f(x+ts) - f(x)}{t} \in \mathbb{R}$$

Theorem 1.6 (Existence of Clark derivative)**.** ***Locally Lipschitz continuous function is Clark differentiable in the point****. And we have:*

$$f^\circ(x, s) \leq L\|s\| \quad \forall s \in \mathbb{R}^n$$

Form convexity follows the local Lipschitz differentiability. Therefore all the convex function is Clark differentiable.

Lemma 1.7. *Given a function $f : D \longrightarrow \mathbb{R}, D \subset \mathbb{R}^n$, locally Lipschitz continuous in the point $\hat{x}$ with Lipschitz constant L. Than the clark derivative $f^\circ(\hat{x}, s)$ is:*

- *sublinear:* $f^\circ(x, u+v) \leq f^\circ(x, u) + f^\circ(x, v)$
- *reflexiv:* $f^\circ(x, -u) = -f^\circ(x, u)$

Theorem 1.8. *Given a convex function: $f : D \longrightarrow \mathbb{R}$ (therefore locally Lipschitz continuous, the local Lipschitz continuity is important!). Directional derivative and Clark derivative are equivalent.*

Proof. Aufgab 3, Set 2. □

Definition 1.4 (Clarke Subdifferential)**.** $f : D \longrightarrow \mathbb{R}, D \subset \mathbb{R}^n$ open, and is local Lipschitz continuous.The Clark derivative is well defined and the Clark sub-differential could be defined as:

$$\partial^C f(x) := \{y \in \mathbb{R}^n \mid f^\circ(x, s) \geq \langle y, s \rangle, \forall s \in \mathbb{R}^n\}$$

Example (Half pipe example)**.** in the point 0 not continuous differentiable. If we want $\partial f(\hat{x}) = \{\nabla f(\hat{x})\}$, the smoothness of f would be required. We see that even for differentiable function the gradient is not necessarily equivalent to Clark sub-derivative.(Unlike sub-derivative)
In general Banach space obvious more complicated.

Theorem 1.9. ***Locally Lipschitz continuous function admit in every point the Clark sub-derivative, and the sub-derivative*** *and:*

1. $\partial^C f \subset U_L(0)$ *is non-empty, convex, and compact.*
2. $f^\circ(x, s) = \max\{y^\top s | y \in \partial^C f(x)\}$ *for all* $s \in \mathbb{R}^n$

Theorem 1.10 (NBO: necessary condition for the optimal point)**.**

1.3 Bouligand Differentiability(Limiting Jacobian)

Theorem 1.11 (Theorem of Rademacher)**.** ***Local Lipschitz continuous functions are a.e. differentiable.***

Definition 1.5 (Bouligand derivative)**.** Let $f : D \longrightarrow \mathbb{R}^m, D \subset \mathbb{R}^n$ open, local Lipschitz continuous in D. D_f the dense subset in the sense of Rademacher. The Bouligand subdifferential(Limiting Jacobian) is defined as:

$$\partial^B f(x) := \{G \in \mathbb{R}^{m \times n} \mid \exists (x_k) \subset D_f, x_k \longrightarrow x, \nabla f \longrightarrow G\}$$

Theorem 1.12. *Let f defined as above. The Clark derivative in any point in D is the convex hull of the Bouligand derivative.:*

$$\partial^C f(x) = conv\{\partial^B f(x)\}$$

Example.

1.4 Optimization condition for the abs-normal functions

We encounter constantly nonsmooth function, let f be a multivariate function defined on a open set. The functions:

- PC^r-function
- Piecewise Smooth function
- Piecewise Linear function

are all nonsmooth.

Example. niemals glatt verfahren auf nicht linear problem zu verwenden.

Definition 1.6 (Abs-normal function)**.** For the piecewise smooth function we define the abs-normal form. Given $d \in \mathbb{N}$, piecewise smooth function$f : \mathbb{R}^n \longrightarrow \mathbb{R}, y = f(x)$ we define

$$z = F(x, |z|)(\text{implizit defined})$$
$$y = \varphi(x, |z|)$$

where $\varphi, F \in C^c_{abs}(\mathbb{R}^n, \mathbb{R}+)$.

Example.

Theorem 1.13 (necessary conditions of optimization for the C^d_{abs} Functions)**.** *Given $f \in C^d_{abs}(\mathbb{R}^n), d \geq 1$. If x_* is a minimal point, $z(x_*) = 0$, $rank(Z) := rank(z(z_*, x_*)) = s$. Then there must exist a vector of Lagrange multiplier with $\lambda_* \in \mathbb{R}^s$, such that:*

- $a^\top(x_*, 0) + \lambda_*^\top Z(x_*, 0) = 0$
- $F(x_*, 0) = 0$
- $b^\top(x_*, 0) + \lambda_* \frac{\partial F}{\partial |z|} = 0$

The evaluation procedure

2 Subgradient Methods

2.1 introduction

gradient ist immer eine abstiegrichtung. Aber sub-gradient nicht.

Definition 2.1 (descent, steepest descent)**.** Given $f : D \longrightarrow \mathbb{R}, D \subset \mathbb{R}^n$ directional differentiable. We define

- The direction $s \in \mathbb{R}^n$ is called descent for f in x, if

$$f'(x, s) < 0$$

- The descent is called the steepest descent if

$$f'(x, \frac{s}{\|s\|}) = \min_{\|d\|=1} f'(x, d)$$

Remark. .

1. $s = 0$ is allowed

2. Is f Lipschitz continuous in x, it can be proved that the directional derivative is also (Lipschitz) continuous. By the theorem of Weierstrass the existence of the minimum is garanteed.

3. Is f differentiable in x, the steepest descent is $s = -\alpha \nabla f(x), \alpha > 0$.

4. By the theorem of Rademacher, for the Lipschitz-continuous function, the points not differentiable is a null set.

Definition 2.2 (sub-gradient, subdifferential)**.** Given $f : D \longrightarrow \mathbb{R}$ convex on a convex, open subset of $\mathbb{R}^n$. A vector is called sub-gradient of f in the point $\hat{x}$, if

$$f(x) - f(\hat{x}) \geq g^\top (x - \hat{x}), \forall x \in D \tag{I.4}$$

The set $\partial f(\hat{x}) \subset \mathbb{R}^n$ with

$$\partial f(\hat{x}) := \{g \in \mathbb{R}^n \mid g \text{ subgradient}\}$$

is called the subdifferential of f in point $\hat{x}$.

2.2 subgradient method

Algorithm 1 subgradient algorithm

$f : \mathbb{R}^n \longrightarrow \mathbb{R}$ directional differentiable for all $x \in \mathbb{R}^n$ and $x^0 \in \mathbb{R}^n$ is given. For $k = 0, 1, 2, \ldots$

```
opt_sub(f, ε, x_0)
s_0 ∈ ∂f(x_0)
while (s_k ≤ ε) do
    Calculate one sub-derivative s^k ∈ ∂f(x^k)
    Determine a step width α_k > 0
    Set d^k = -s^k/||s||,   x^{k+1} = x^k + α_k d^k,   k = k+1
end while
```

Remark.

1. We would expect the one sub-gradient $g \in \partial f(x)$ can be calculated in every step $\Rightarrow$ Oracle.

2. The sub-gradient method is not a descent method.

Lemma 2.1. *$f : \mathbb{R}^n \longrightarrow \mathbb{R}$ a convex function. $\tilde{x}$ an arbitrary minimal point of f and x^k the k-th iteration of the sub-gradient method and not the minimal point. Then there exists $\tilde{\alpha} > 0$, s.t.for $x^{k+1} = x^k + \alpha_k d^k$ we have:*

$$\|x^{k+1} - \tilde{x}\| < \|x^k - \tilde{x}\| \quad \forall \alpha_k \in (0, \tilde{\alpha})$$

Proof. By the sub-gradient method we get:

$$\|x^{k+1} - \tilde{x}\| = \|x^k - \tilde{x}\|^2 + 2\alpha_k \langle d_k, x^k - \tilde{x}\rangle + \alpha_k^2$$

□

Lemma 2.2. *$f : \mathbb{R}^n \longrightarrow \mathbb{R}$ convex function, $\tilde{x}, z \in \mathbb{R}^n$ with $f(z) > f(\tilde{x})$ and $s \in \partial f(z), s \neq 0$. Define $H_z = \{x \in \mathbb{R}^n | \langle s, x - z\rangle = 0\}$. Then the optimisation problem*

$$\min_{x \in H_z} \frac{1}{2} \|x - \tilde{x}\|^2$$

has an unique deterministic solution:

$$x^* = \tilde{x} + \lambda s, \text{ with } \lambda = \frac{\langle s, z - \tilde{x}\rangle}{\|s\|^2}$$

And we have $\lambda > 0, \|x^* - \tilde{x}\| = \lambda \|s\|$, and $f(x^*) \geq f(z)$

Proof. The objective function is convex and differentiable. And we have an equality constraint.

1. We apply the Lagrange multiplier:

$$\begin{aligned}\nabla f(x^*) - \lambda s &= 0\\ \langle s, x^* - z\rangle &= 0\end{aligned}$$

   solve for the λ.

2. Since $f(z) > f(\tilde{x})$ we have:

$$0 \geq f(\tilde{x}) - f(z) \leq \langle s, x - z\rangle = -\lambda \|s\|^2$$

   So we have that $\lambda > 0$.

3. By the definition of the sub-gradient we have:

$$f(x^*) - f(z) \geq \langle s, x^* - z\rangle = 0$$

□

Remark.

1. We mainly apply the Lagrange and the definition of the sub-gradient.

2. Notice that the sub-gradient method is not a descent method.

Theorem 2.3. *$f : \mathbb{R}^n \longrightarrow \mathbb{R}$ the objective function, which is **convex** with $\inf_{x\in\mathbb{R}^n} f(x) := f^* > -\infty$. If the step width $\{\alpha_k\}_{k\in\mathbb{N}}$ satisfies:*

$$\sum \alpha_k = \infty, \quad \lim \alpha_k = 0$$

then the sub-gradient method will not stop in finite steps but converges to the minimal point:

$$\lim f_k \longrightarrow f^*$$

2.3 proximal algorithm

Definition 2.3 (the proximal operator)**.** Given $f : \mathbb{R}^n \to \mathbb{R}$ a proper closed convex function, the proximal operator is defined as

$$\text{prox}_f : \mathbb{R}^n \to \mathbb{R}^n, \quad \text{prox}_f(x) := \arg\min_{y\in\mathbb{R}^n} \left(f(y) + \frac{1}{2}\|x - y\|^2 \right).$$

The proximal operator is the generalization of the projection operator, indeed, consider f the indicator function of the closed convex subset C, then $\text{prox}_f(x)$ is the orthogonal projection of x on C.

Theorem 2.4 (fixed point property)**.** *x^* is a minimizer of f (proper closed and convex) if and only if*

$$\text{prox}_f(x^*) = x^*.$$

Proof.
"$\Rightarrow$": $x^* \in \arg\min f$, then

$$\text{prox}_f(x^*) = \arg\min_y (f(y) + \frac{1}{2}\|y - x^*\|^2).$$

It holds

$$f(x^*) + \frac{1}{2}\|x^* - x^*\| \leq f(y) + \frac{1}{2}\|y - x^*\|^2, \quad \forall y \in \mathbb{R}^n.$$

Hence $\text{prox}_f(x^*) = x^*$.
"$\Leftarrow$": assuming $x^* = \text{prox}_f(x^*)$, then

$$0 \in \partial f(x^*) + |x^* - x^*|,$$

hence

$$0 \in \partial f(x^*).$$

□

Definition 2.4 (firmly nonexpansive operator)**.** An operator A is firmly nonexpansive if

$$\forall x, y: \quad \|A(x) - A(y)\|^2 \leq \langle x - y, A(x) - A(y)\rangle.$$

Definition 2.5 (Moreau decomposition)**.** Given f a proper closed convex function, it holds

$$\forall v \in v = \mathbb{R}^n: \quad \text{prox}_f(v) + (\text{prox}_f)^*(v).$$

Lemma 2.5. *Given I_L the indicator function for L subspace:*

$$I_L(x) = \begin{cases} 0 & x \in L \\ +\infty & x \notin L, \end{cases}$$

it holds

$$(I_L)^* = I_{L^\perp}.$$

Corollary 2.5.1 (orthogonal decomposition)**.** *Given L a subspace, it holds*

$$\forall v \in \mathbb{R}^n: \quad v = \Pi_L v + \Pi_{L^\perp} v.$$

Interpretation

Definition 2.6 (infimal convolution)**.** Given 2 proper closed convex function f, and g, the infimal convolution is defined as

$$f \square g(v) := \inf_x \left(f(x) + g(v - x) \right).$$

Lemma 2.6. *It holds*

1. $(f \square g)^* = f^* + g^*$

Definition 2.7 (the Moreau envelop)**.** Given the convex proper closed function f and $\lambda > 0$, the Moreau envelop is defined as

$$M_{\lambda f}(v) := \inf_{x \in \mathbb{R}^n} \left(f(x) + \frac{1}{2\lambda}\|x - v\|^2 \right).$$

is the Moreau envelope of f with the parameter λ.

Lemma 2.7. *It holds*

1. *M_f is differentiable,*

2. $\mathrm{dom}(M_f) = \mathbb{R}^n$,

3. $\arg\min_x M_f \equiv \arg\min_x f$,

4. $M_f^{**} = M_f$.

Corollary 2.7.1. *It holds*

$$M_f = M_f^{**} = (f^* + \|\cdot\|_2^2)^*.$$

Theorem 2.8. *The proximal operator returns to the infimizer of M_f, i.e.,*

$$M_f(x) = f(\mathrm{prox}_f(x)) + \frac{1}{2}\|x - \mathrm{prox}_f(x)\|_2^2.$$

moreover,

$$\mathrm{prox}_{\lambda f}(x) = x - \nabla M_{\lambda f}(x)$$

Proof. □

Theorem 2.9. *It holds*

$$\mathrm{prox}_{\lambda f}(x) = (I + \lambda \partial f)^{-1}.$$

$(I + \lambda \partial f)^{-1}$ *is called the resolvent.*

Proof. □

Corollary 2.9.1 (proximal operator as modified gradient step)**.** *Assuming f is twice differentiable, for $\lambda > 0$ sufficiently small, it holds*

$$\mathrm{prox}_{\lambda f}(x) = x - \lambda \nabla f(x) + o(\lambda).$$

Proof. □

2.4 cutting-plane methods

The cutting-plane method is motivated by the subdifferential. The subdifferential at each point of a function defines a linear hyperspace and a minorant of the original function. we show that the minimax of a family of functions could be transformed to a linear optimisation problem. Theorem 3.13 shows that this method indeed converges to the minimum of the function. Then 3.14 provides a valid termination criteria by stating that if by k-th iteration the approximated value is in the ε-ball of the real value, then the approximation is within the ε-ball of the minimum of the objective function.

By the definition of sub-gradient we know that $f(x)$ is lower bounded by the function:

$$f_k^{se} = \max_{1 \le j \le k} l_j(x) = \max_{1 \le j \le k} \{f(x_j) + s_j^\top (x - x_j)\} \tag{I.5}$$

The next iteration x_{j+1} is determined by the function:

$$\min_{x \in D} f_k^{se}(x)$$

The advantage is that we now have a **smooth** optimisation problem:

$$\min_{x \in D, \alpha \in \mathbb{R}^n} \alpha \quad \text{such that} \quad f_j^{se}(x) - \alpha \le 0 \quad 0 \le j \le k \tag{I.6}$$

In the case D is convex we can transform it to a linear optimisation problem and it is analog easy to solve.

Lemma 2.10. *For functions $f_j : D \longrightarrow \mathbb{R}, j = 1, \cdots, k$, with $D \subset \mathbb{R}^n$, we observe the optimisation problem:*

$$\min_{x \in D} f(x) \qquad f(x) = \max_j f_j(x) \tag{I.7}$$

$$\min_{a \in \mathbb{R}} a \qquad f_j(x) \leq a, x \in D, 1 \leq j \leq k \tag{I.8}$$

The two conditions are equivalent.

Given $f : \mathbb{R}^n \longrightarrow \mathbb{R}$ for all $x \in \mathbb{R}^n$ directional differentiable and $x_0 \in \mathbb{R}^n$ given. For $k = 1, 2, \ldots,$

1. calculate a sub-gradient for each point $g^k \in \partial f(x^k)$
2. Terminate when conditions satisfied
3. Calculate x^{k+1} by solving the problem (8)

Theorem 2.11. *Let $D \subset \mathbb{R}^n$ be closed, $f : D \longrightarrow \mathbb{R}$ Lipschitz-continuous and convex and the Algorithm 3.12 produced an unlimited sequence. Then every cluster point of the sequence is an optimal solution of*

$$\min_{x \in D} f(x)$$

Proof. idea: By showing that the minorant f_k^{se} converges to a point in f. □

Theorem 2.12. *Given $f : D \longrightarrow \mathbb{R}^n$ a continuous and convex function. The sequence generated by the algorithm 3.12 has a cluster point $\overline{x}$. By theorem 3.13 we know that the cluster point is an optimal solution of the objective function $\min_{x \in D} f(x)$. If we have:*

$$0 \leq f(x^{k+1}) - f_k^{se}(x^{k+1}) \leq \varepsilon$$

for an $\varepsilon \geq 0$, Then we have:

$$0 \leq f(\overline{x}) - f(x^{k+1}) \leq \varepsilon$$

By the above theorem we have termination criteria. Moreover, we observe that the Schnittebenenverfaheren is a descendant method.

However, the problem (8) is not guaranteed to be solvable. Consider $D = \mathbb{R}$ and $s_0 \neq 0$. Then l_0 is not lower bounded. This problem could be dealt with by penalty term.

3 The Bundle-Methods

3.1 Bundle-method motivated by Schnittebeneverfahren

Given the objective function , observe the sub-problem:

$$\min_{d \in \mathbb{R}^n} f_{J_k}^{se}(x + d) + \frac{1}{2\gamma} \|d\|^2 \tag{I.9}$$

Lemma 3.1. *Let d^k be a solution of* **??** *, then there exists a $\Delta_k > 0$, such that d^k solves the Trust-Region-Problem*

$$\min_{d \in \mathbb{R}^n} f_{J_k}^{se}(x^k + d), s.t \quad \|d\| \leq \Delta_k$$

On the other hand if d^k solves the Trust-Region-Problem, and the Nebenbedingung $\|d\| \leq \Delta_k$ is strongly active, then there exist an $\gamma_k > 0$, such that d^k the penalised Schnittebene problem solves.

We transform the objective function in the neighbourhood of x_k:

$$f_{J_k}^{se}(x) = \max_{j\in J_k} l_j = \max_{j\in J_k}(s^j)^T(x-x^j)+f(x_j)$$
$$= f(x^k)+\max_{d\in\mathbb{R}^n}((s^j)^T(x-x^k)-\alpha_j^k) := f(x^k)+\overline{f_{J_k}^{se}}.$$

where

$$\alpha_j^k = f(x^k)-l_j(x^k)\geq 0$$

observe now the penalised objective function:

$$\min_{d\in\mathbb{R}^n}\overline{f_{J_k}^{se}}(x_k+d)+\frac{1}{2\gamma_k}\|d\|^2$$

is the error of j-th iteration evaluated at x_k. By lemma 3.11 the optimisation problem becomes:

$$\min_{d\in\mathbb{R}^n,\xi\in\mathbb{R}}\xi+\frac{\|d\|^2}{2\gamma_k}, \qquad (s^j)^Td-\alpha_k^j-\xi\leq \qquad \forall j\in J_k \tag{I.10}$$

The pair (d^k,ξ^k) is the solution of the problem.

Lemma 3.2. (d^k,ξ^k) *is the solution of* **??** *if and only if the KKT conditions is satisfied:*

pred vs. ared

For a step d^k we compute if the iteration really reduces the value of the function. We define:

$$ared(d^k) = f(x^k)-f(x^k+d^k)\geq 0$$

for k large eenough, we have the predicted reduction:

$$pred(x_k) = f_{J_k}^{se}(x_k)-f_{J_k}^{se}(x_k+d^k) = -\xi^k$$

We have then the concept of serious step and zero step.

Given : $f:\mathbb{R}^n\to\mathbb{R}$, $\gamma_+,\gamma_->0$.

1. choose $\gamma\in[\gamma_-,\gamma_+]$, and calculate the KKT condition : (d^k,λ^k,ξ^k)
2. Calculate $v^k=-\frac{d_k}{\gamma_k}$ and $\varepsilon_k=\sum_{j\in J_k}\lambda_j\alpha_j^k$ (linearised error)
3. Termination criteria: $\|v_k\|\leq\varepsilon$, or $\varepsilon_k\leq\varepsilon$.
4.

3.2 Optimisation method with the ε-subdifferential

Definition 3.1. Given $f:\mathbb{R}^n\to\mathbb{R}$ convex and $\varepsilon\geq 0$. The vector $s\in\mathbb{R}^n$ is called ε-subgradient of f in x, if :

$$f(y)-f(x)+\varepsilon\geq s^T(y-x) \quad \forall y\in\mathbb{R}^n$$

and the set of ε-subdifferential is defined as:

$$\partial_\varepsilon f(x) := \{s\in\mathbb{R}^n | \text{s } \varepsilon\text{-subdifferential of } f \text{ in } x\}$$

Alternatively we have the equivalent condition:

$$l(x) := f(\hat{x})+g^T(x-\hat{x})-\varepsilon\leq f(x)\Leftrightarrow g\in\partial_\varepsilon f(x)$$

Definition 3.2 (ε-directional derivative)**.** Given a convex function from $\mathbb{R}^n$ to $\mathbb{R}$.

$$f'_\varepsilon(x,s) = \inf_{t>0} \frac{f(x+ts) - f(x) - \varepsilon}{t}$$

Theorem 3.3. *$f : \mathbb{R}^n \to \mathbb{R}$, $x \in \mathbb{R}$:*

1. $\partial_\varepsilon f(x) = \{s \in \mathbb{R}^n | s^T v \leq f_\varepsilon(x,v) \forall v \in \mathbb{R}^n\}$
2. $\partial_\varepsilon f(x)$ *is a convex, nonempty and compact.*
3. $f'_\varepsilon(x,v) = \max_{s \in \partial_\varepsilon f(x)} s^T v \qquad \forall v \in \mathbb{R}^n$

Theorem 3.4. *Given $f : \mathbb{R}^n \to \mathbb{R}$ convex*

1. $\overline{x}$ *is ε- optimal, i.e.*

$$f(\overline{x}) \leq f(x) + \varepsilon \qquad \forall x \in \mathbb{R}^n$$

2. $f'_\varepsilon(x,v) \geq 0$, $\forall v \in \mathbb{R}^n$
3. $0 \in \partial_\varepsilon f(\overline{x})$

Theorem 3.5. *Given $f : \mathbb{R}^n \to \mathbb{R}$ convex, then*

1. $\forall \varepsilon > 0 \quad \exists \delta > 0$*:*

$$\cup_{y \in U_\delta(x)} \partial f(y) \subset \partial_\varepsilon f(x)$$

2. $\forall \delta > 0 \quad \exists \varepsilon > 0$*:*

$$\cup_{y \in U_\delta(x)} \partial f(y) \subset \partial_\varepsilon f(x)$$

The definition of projector: P is a projector if:

$$P \circ P = P$$

Observe that:

$$P(x) = \arg\min_{y \in Y} \|x - y\|$$

is a projector.

Theorem 3.6. *Given $f : \mathbb{R}^n \to \mathbb{R}$ convex, $\varepsilon \geq 0$ and $x \in \mathbb{R}^n$ with $0 \notin \partial_\varepsilon f(x)$. For $\tilde{d} = -s$ with $s = P_{\partial_\varepsilon f(x)}(0)$, $d = \frac{\tilde{d}}{\|\tilde{d}\|}$ is a solution for:*

$$\min_{\|v\|=1} f'_\varepsilon(x,v)$$

and we have:

$$f'_\varepsilon(x,\tilde{d}) = -\|\tilde{d}\|^2 \leq 0$$

$$\inf_{t>0} f(x + t\tilde{d}) < f(x) - \varepsilon$$

Theorem 3.7.

Algorithm 2 ε-subdifferential algorithm

Given $f : \mathbb{R}^n \to \mathbb{R}$ directional differentiable $\forall x \in D$.

1. Calculate: $s^k = P_{\alpha_\varepsilon f(x^k)}(0)$
2. Falls $s^k = 0$ STOP;
3. Plug $d_k = -s_k$ and calculate the optimal step α_k:

$$f(x_k + \alpha_k d_k) = \min_{\alpha > 0} f(x_k + \alpha d_k)$$

4. set $x_{k+1} = x_k + \alpha_k d_k$

4 Techniques

4.1 Convex functions

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y) \tag{I.11}$$

$$f(a) + f(b) \leq f(a+b) \tag{I.12}$$

$$f(x + h(\tilde{x} - x)) \leq h f(\tilde{x}) + (1-h) f(x) \tag{I.13}$$

$$f(\tilde{x}) - f(x) \geq \frac{f(x + h(\tilde{x} - x)) - f(x)}{h} = \varphi(h) \tag{I.14}$$

Chapter II

Machine learning

1 Introduction

1.1 The First Question: What is machine learning?

A possibile definition.

Definition 1.1. A program is said to learn from the experience **E** w.r.t. some task class **T** and performance measure **P**, if its performance at tasks in `T` is measured by **P** improves with experience **E**.

This course: How to determine the mathematical optimisation method for the tasks.

1.2 Examples

LSQ problem

Deep Learning

The underlying model: Neurone Network. The net structure is determined by the activation function and propagation weights, the parameter involves:

- # of layers
- # of neurones pro layer
- activation functions
- the weights

Given the net structure and activation function we observe:

$$\min F(\underbrace{x, y}_{\texttt{labeled date}}) = \sum_i \underbrace{l}_{\texttt{loss function}} (y(x_i), y_i) \tag{II.1}$$

where i denote the number of data.

Example (MNIST)**.** Activation function: the soft-max:

$$G : \mathbb{R}^k \to \mathbb{R}^k, \quad z_i \mapsto \frac{e^{z_i}}{\sum_{j=1}^k e^{z_j}} \tag{II.2}$$

The soft-max minimises the `argmax` function and is often used in the final layer for classification

Example (Other activation functions)**.**

Rectified Learning Unit:

$$\rho(y_i) = \max(y_i, 0) \tag{II.3}$$

Remark: not differentiable, works well in application

Sigmoid function

$$\rho(x) = \frac{1}{1+e^{-x}} \tag{II.4}$$

Remark: Differentiable, but little gradient information for $|x|$ large.

The Deep Learning method the mathematically well researched, we have the following theorem:

Theorem 1.1 (Yarotsky)**.** *Every $f \in W^{n,\infty}([0,1]^d)$ can be approximated by a neurone network with maximal $c \cdot (\ln(\frac{1}{\varepsilon}+1))$ layers and $c \cdot \varepsilon^{\frac{d}{n}}(\ln(\frac{1}{\varepsilon}+1))$ weights.*

1.3 Properties of machine learning

Central Requirement: The optimised model can be generalised.

Procedures:

1. Separation of Data.

2. Train the model.

3. Test the model with other data.

The measurement of quality:

1. small training error

2. small gap between training error and generalisation error.

Large gap leads to *overfitting* if the training error is too small w.r.t. the generalisation error and *underfitting* if the training error is too large.

1.4 Mathematical Optimisation

General formulation:

$$\min_{x \in X} F(x) \quad X \subset \mathbb{R}^n \texttt{ some feasible set}$$

`where`

$$F : \mathbb{R}^n \to \mathbb{R}$$

The Gradient Method

Remark. Note that all the following quantities can be used as stop criteria:

1. $\|\nabla f(x)\|$
2. $\|f(x_{k+1}) - f(x_k)\|$
3. $\|x_{k+1} - x_k\|$

Remark. The relative criteria is invariant under scaling of the cost function(robust against scalaring).

Algorithm 3 gradient method

Given $f := \mathbb{R}^n \to \mathbb{R}$ and $\varepsilon > 0$ a maxiter.

`For` $k = 0, \cdots : f(x_{k+1}) = f(x_k) - \alpha_k \nabla f(x_k)$

if :

$\|\nabla f(x_k)\| \leq \varepsilon$ **STOP** `(absolute stop criterium)`

$\|\nabla f(x_k)\| \leq \varepsilon \|\nabla f(x_0)\|$ **STOP** `(relative stop criteria)`

$k \geq$ `maxiter`:**STOP**

else:

$d_k = -\nabla f(x_k)$

`find` α_k `s.t.` $f(x_k + \alpha_k d_k) < f(x_k)$

$x_{k+1} = x_k + \alpha_k d_k$

Method of the step size selection

The following method available:

1. **Backtracking:** choose $\alpha_k \equiv 1$, if not a descendent, reduce it. REMARK: efficient, not convergence result.
2. **Wolf-Powell:**
3. In machine learning, α_k is usually called the learning rate, which is chosen as a constant.

Theorem 1.2 (Convergence of the Gradient Method)**.** *For $f \in C^1(\mathcal{D}, \mathbb{R}^n)$, we have*

1. *The method converges to a stationary point.*
2. *For f bounded from below,* $\lim_{k\to\infty} \nabla f(x_k) = 0$.

Other method of the optimisation available such as Newton Method, Quasi-Newton Method.

Definition 1.2 (q- and r-linear convergence)**.** Given $(v_k)_{k\in\mathbb{N}} \subset \mathbb{R}_+$ with

$$\lim_k v_k \to 0$$

we say v_k converges

$$\begin{aligned} &\textit{q-linear (quotient)} \ \exists c \in [0,1) : v_{k+1} \leq c v_k \quad \forall k = 0, \ldots \\ &\textit{r-linear (root)} \qquad \exists c \in [0,1) : v_k \leq c^k M \quad \forall k = 0, \ldots \end{aligned} \tag{II.5}$$

for some $M > 0$.

Remark.

1. The q-linear implies the r-linear, the inverse doesn't hold.
2. q-linear is a strong property.

Algorithm 4 Gradient Method with Constant Step Size

$x_0 \in \mathcal{D}$, calculate $\nabla f(x_0)$ $\|\nabla f(x_k)\| > \varepsilon$ **or** $\|\nabla f(x_k)\| > \varepsilon \|f(x_k)\|$ $x_{k+1} = x_k - \frac{1}{L}\nabla f(x_k)$

Gradient Method with constant step size

the constant L is in general **unknown**. We will see in this section for $f \in C^1(\mathcal{D}, \mathbb{R}^n)$ with $\nabla f \in Lip(\mathcal{D}, \mathbb{R}^n)$, we may choose $L = Lip(\nabla f)$. Question now is the convergence rate of the method.

Definition 1.3 (sublinear convergence)**.** Given a vanishing sequence as above, we list the following criteria for sub-linear convergence, if there exists some $\bar{k}$ s.t. $\forall k \geq \bar{k}$:

1. $v_k \leq \frac{M}{\sqrt{k}}$
2. $v_k \leq \frac{M}{k}$
3. $v_k \leq \frac{M}{k^2}$

Theorem 1.3 (sub-linear convergence rate for constant step-size)**.** *Given $f \in C^1(\mathbb{R}^n, \mathbb{R})$, $\nabla f \in Lip(\mathbb{R}^n, \mathbb{R})$ with Lipschitz constant L, then the algorithm is sublinear convergent with*

$$\min_{0 \leq k \leq K} \|\nabla f(x_k)\| \leq \frac{\sqrt{2L(f(x_0) - f(x_*))}}{\sqrt{K}} \tag{II.6}$$

Proof. By the assumption we have by Taylor:

$$f(x_{k+1}) \leq f(x_k) + \langle \nabla f(x_k), \alpha_k d_k \rangle + \frac{L}{2}\alpha_k^2 \|d_k\|^2 \tag{II.7}$$

By the assumption of the constant step size $\alpha_k = \frac{1}{L}$ we have

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2 \quad \texttt{or} \ \|\nabla f(x)\| \leq 2L\Big(f(x) - f(y)\Big). \tag{II.8}$$

By induction we have

$$\min_{0 \leq j < K} \|\nabla f(x_j)\| \leq \sqrt{\frac{\sum_{j=0}^{K-1} \|\nabla f(x_j)\|^2}{K}} \overset{\texttt{induction}}{\leq} \sqrt{\frac{2L(f(x_0) - f(x_*))}{K}}.$$

□

Summary of CGM:

1. **Positive:** simple
2. **Negative:** Information of L required, which is often difficult to get through algorithmic differentiation.

Instead of the Lipschitz continuity of the gradient, we now introduce another condition, which is easier to verify.

Definition 1.4 (Convexity and Strong convexity)**.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is **convex** if

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle. \quad \forall x, y \tag{II.9}$$

A function $f : \mathbb{R}^n \to \mathbb{R}$ is **strongly convex** if there exists $C \geq 0$, such that

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{C}{2}\|y - x\|^2 \quad \forall x, y \in \mathbb{R}^n \tag{II.10}$$

Remark.

1. Strong convexity implies the *strict convexity,* hence the existence of a unique minimiser.

2. If additionally, $\nabla f \in Lip(\mathbb{R}^n, \mathbb{R})$, then we have $C < L$.

Theorem 1.4 (Convex case (p30))**.** *For the convex function f the steepest descent method with the constant steplength $\alpha_k = \frac{1}{L}$ converges sublinearly with*

$$f(x_{k+1}) - f_* \leq \frac{L}{2K}\|x_0 - x_*\|^2. \tag{II.11}$$

Proof. By convexity we have

$$f(x_*) \geq f(x_k) + \langle \nabla f(x_k), x^* - x^k \rangle \tag{II.12}$$

By the ness we have

$$f(x_{k+1}) - f(x_k) \leq -\frac{1}{2L}\|\nabla f(x_k)\|^2$$

Plug in we have

$$f(x_{k+1}) - f(x_*) \leq \langle \nabla f(x_k), x_k - x_* \rangle - \frac{1}{2L}\|\nabla f(x_k)\|^2$$

hence,

$$f(x_{k+1}) - f(x_*) \ \leq \frac{L}{2}\left(\|x_k - x_*\|^2 - \|x_{k+1} - x_*\|^2\right). \tag{II.13}$$

By induction we have the convergence rate stated in (II.11). □

Theorem 1.5 (q-linear convergence of CGM under strong convexity)**.** *Given a strongly convex with constant c and L-smooth with constant L, the Algorithm* **??** *with $\alpha_k = \frac{1}{L}$ converges* ***q-linearly*** *with*

$$f(x_{k+1}) - f(x_*) \leq \left(1 - \frac{c}{L}\right)(f(x_k) - f(x_*)) \tag{II.14}$$

Proof. By Lipschitz continuity of the gradient we get as (II.8)

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2$$

By the strong convexity we get

$$f(y) - f(x) \geq \langle \nabla f(x), y - x \rangle + \frac{c}{2}\|y - x\|^2, \tag{II.15}$$

choose $y = x - \frac{\nabla f(x)}{c}$ and minimise over y we get

$$\begin{aligned} & f(y) - f(x) \geq -\frac{1}{2c}\|\nabla f(x)\|^2 \\ & \Rightarrow \\ & \|\nabla f(x)\|^2 \geq 2c\left(f(x) - f(y)\right) \end{aligned} \tag{II.16}$$

Plug (II.16) into the equation for *L-smooth* functions (II.8) for $x = x_k, y = x_*$ we get

$$\begin{aligned} & f(x_{k+1}) \leq f(x_k) - \frac{c}{L}\left(f(x_*) - f(x_k)\right) \\ & \Rightarrow \\ & f(x_{k+1}) - f_* \leq \left(1 - \frac{c}{L}\right)\left(f(x_k) - f(x_*)\right) \end{aligned}$$

□

Summary of CGM under strong convexity and L-smoothness

1. It's *q-linear* convergent with $c = 1 - \frac{C}{L}$.
2. It's *r-linear* convergent with $c = 1 - \frac{C}{L}$ and $v_0 = f(x_0) - f(x_*)$.
3. Better result can hardly be achieved even by much more sophisticated choice of step size.

1.5 Mathematic Optimisation for ML

Particular difficulties:

1. Large data set: such as image recognition.
2. Sparse data set: such as recommendation system.
3. Large dimensional optimiser: Deep Learning.

2 Gradient Method with Momentum

2.1 Motivation

The momentum method arises in the context that, in the case of pathological curvature, the constant step size method is often inefficient, in particular, a high learning rate may accelerate the process in the low curvature but make the state oscillate in high curvature. One way to adapt the curvature is to incorporate the information of *2nd momentum*.

Observe the dynamic system:

$$\frac{dx}{dt} = -\nabla f(x) \tag{II.17}$$

Notice the equilibrium is a fix point of f, i.e.,

$$x \xrightarrow{t \to \infty} f(x)$$

Analog we have for the dynamic system

$$m\frac{d^2x}{d^2t} = -\nabla f(x) - b\frac{dx}{dt} \tag{II.18}$$

where m denotes the mass and b the fraction constant. We have the **Chebyshev iterative method:**

$$x_{k+1} = x_k - \alpha \nabla f(x_k) + \beta (x_k - x_{k-1}).$$

discretise the system we get the **heavy ball method**

$m_0 = 0.$ $\|\nabla f(x_k) \geq \varepsilon$ `OR` $\|\nabla f(x_k) \geq \varepsilon \|\nabla f(x_0)\|$ $m_{k+1} = \beta m_k - \alpha \nabla f(x_k)$ $x_{k+1} = x_k + m_{k+1}$

The term "heavy ball" is equivalent to acceleration in when $\|\nabla f(x)\|$ is large. Despite it's well established efficiency in practice, it has been proved that there exits **convex** functions, for which the method doesn't converge.

An alternative is proposed by Nesteror, where we evaluate the gradient little towards the **momentum**, in practice it's always faster than heavy ball method numerically. **Nesterov Accelerated Gradient (NAG) method**:

$$\begin{aligned} m_{k+1} &= \beta m_k - \alpha \nabla f(x_k + \beta m_k) \\ x_{k+1} &= x_k + m_{k+1} \end{aligned} \tag{II.19}$$

The Questions:

1. When does the NAG converges(for which α and β).

2. Does it converge to the minimum.

2.2 Nesteror acceralted gradient method

1. For weakly convex functions

$x_0 = y_0, x_{-1} = x_0.$ $\|\nabla f(x_k)\| \leq \varepsilon$ $y_k = x_k + \beta_k(x_k - x_{k-1})$ $x_{k+1} = y_k - \alpha_k \nabla f(y_k)$

with the parameter determined by

$$\begin{aligned}
&\alpha_k = \frac{1}{L} \\
&\lambda_0 = 0 \\
&\lambda_{k+1} = \frac{1+\sqrt{1+4\lambda_k^2}}{2} \quad (\lambda_{k+1} = \lambda_{k+1}^2 - \lambda_k^2) \geq 1 \\
&\beta_k = \frac{\lambda_k - 1}{\lambda_k + 1}
\end{aligned} \tag{II.20}$$

Theorem 2.1 (convergence rate of NAG for convex functions)**.** *Given $f : \mathbb{R}^n \to \mathbb{R}$ convex and L-smooth with Lipschitz constant L, the function admits a minimiser*

$$f_* = f(x_*) = \min_{x \in \mathbb{R}^n} f(x)$$

*Moreover, $\forall x_0 \in \mathbb{R}^n$, the **NGM** converges **sub-linearly** with*

$$f(x_k) - f_* \leq \frac{2L\|x_0 - x_*\|}{(k+1)^2} \tag{II.21}$$

Proof. By L-smoothness (II.8) and convexity we have

$$\begin{aligned}
f\left(y - \frac{1}{L}\nabla f(y)\right) - f(x) &= f\left(y - \frac{1}{L}\nabla f(y)\right) - f(y) + f(y) - f(x) \\
&\leq -\frac{1}{2L}\|\nabla f(y)\|^2 + \langle \nabla f(y), y - x\rangle \\
&= -\frac{1}{2L}\|\nabla f(y)\|_2^2 + \langle \nabla f(y), y - x).
\end{aligned}$$

Plug in $y = y_k, x = x_k$ we have

$$f(x_{k+1}) - f(x_k) \leq -\frac{1}{2L}\|\nabla f(y_k)\|^2 + L\langle x_{k+1} - y_k, y_k - x_k\rangle.$$

Plug in $x_k = x_*$ we have

$$f(x_{k+1}) - f(x_*) \leq -\frac{1}{2L}\|\nabla f(y_k)\|^2 + L\langle x_{k+1} - y_k, y_k - x_*\rangle.$$

With $\lambda_{k+1} = 1$ and define $\delta_k := f(x_k) - f(x_*)$, $u_k = \lambda_{k+1}y_k - (\lambda_{k+1} - 1)x_k - x_*$ we get

$$\lambda_{k+1}^2\delta_{k+1} - \lambda_k^2\delta_k \leq -\frac{L}{2}(\|u_{k+1}\|^2 - \|u_k\|^2).$$

By induction we show

$$\lambda_k^2\delta_k \leq \frac{L}{2}(\|u_0\|^2 - \|u_k\|^2) \leq \frac{L}{2}\|u_0\|^2 = \frac{L}{2}\|x_0 - x_*\|^2$$

□

Example (Best convergence rate for convex functions)**.** For a quadratic problem

$$f(x) = \frac{1}{2}\langle Ax, x\rangle + e_1 x$$

`with`

$$A = \begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & -1 & \dots & 0 \\ 0 & -1 & 2 & -1 & \dots \\ \dots & & & & \\ \dots & \dots & -1 & 2 & -1 \\ \dots & \dots & \dots & 0 & 0 \end{bmatrix}$$

The function admits a minimiser

$$(x_*)_i = 1 - \frac{i}{n+1}$$

The quadratic function is convex with Lipschitz continuous gradient. Since A is positive semi-definite but not positive definite, the function is not strongly convex. For $x_0 = 0$, observe the first optimality condition

$$\nabla f(x) = Ax + e_1$$

Iterative we see that the k-th iterate only has the first k non-negative entries, hence

$$\|x_k - x_*\|^2 \geq \sum_{j=k+1}^{n} (1 - \frac{j}{n+1})^2$$

hence

$$f(x_k) - f(x_*) \geq \frac{M}{(k+1)^2}$$

2. Strongly convex functions

Example (Under strong convexity *r-linear* rate can be achieved)**.** For a quadratic problem

$$f(x) = \frac{1}{2}\langle Ax, x\rangle - b^T x + c$$

`with`

$$A = \begin{bmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & -1 & \dots & 0 \\ 0 & -1 & 2 & -1 & \dots \\ \dots & & & & \\ \dots & \dots & -1 & 2 & -1 \\ \dots & \dots & \dots & -1 & 2 \end{bmatrix}$$

The 1st optimality condition gives the minimiser:

$$x_* = A^{-1} b$$

By the **NAG** method we have

$$m_{k+1} = \beta m_k - \alpha \nabla f(x_k + \beta m_k)$$

$$x_{k+1} = x_k + m_{k+1}$$

Subtract both side by x_* and get the matrix formulation:

$$\begin{bmatrix} x_{k+1} - x_* \\ x_k - x_* \end{bmatrix} = \underbrace{\begin{bmatrix} (1+\beta)(I - \alpha A) & -\beta(I - \alpha A) \\ I & 0 \end{bmatrix}}_{:=W} \begin{bmatrix} x_k - x_* \\ x_{k-1} - x_* \end{bmatrix} \tag{II.22}$$

Definition 2.1 (The spectral radius)**.** The spectral radius of a matrix W is defined as

$$\rho(W) := \{|\lambda| : \lambda \texttt{ is a EW of } W\} \tag{II.23}$$

we define accordingly the operator norm

$$\|W\| := \rho(W) \tag{II.24}$$

Lemma 2.2. *Given the quadratic function with A positive definite with the eigenvalue:*

$$0 < m = \lambda_1 \leq \lambda_2 \cdots \leq \lambda_m = M$$

Then the matrix W has the complex eigenvector:

$$u_{j,1,2} = \frac{1}{2}\left((1+\beta)(1-\alpha\lambda_j) \pm i\sqrt{4\beta(1-\alpha\lambda_j) - (1+\beta)^2(1-\alpha\lambda_j)^2}\right). \tag{II.25}$$

Theorem 2.3 (r-linear CR for strongly convex quadratic functions)**.** *Given a quadratic function with the eigenvalues m and M. Set*

$$\begin{aligned} \alpha &= \frac{1}{M} \\ \beta &= \frac{\sqrt{M}-\sqrt{m}}{\sqrt{M}+\sqrt{m}} = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1} \\ &\texttt{with} \\ \kappa &= \frac{M}{m} > 1 \end{aligned} \tag{II.26}$$

*Then the **NAG** method converges **q-linearly** with*

$$\rho(W) \leq 1 - \frac{1}{\sqrt{\kappa}} < 1 \tag{II.27}$$

with W defined as

$$\begin{bmatrix} x_{k+1} - x_* \\ x_k - x_* \end{bmatrix} = \underbrace{\begin{bmatrix} (1+\beta)(I-\alpha A) & -\beta(I-\alpha A) \\ I & 0 \end{bmatrix}}_{:=W} \begin{bmatrix} x_k - x_* \\ x_{k-1} - x_* \end{bmatrix} \tag{II.28}$$

Proof. By the eigenvalue decomposition the matrix symmetric matrix A can be decomposed as

$$A = U\Lambda U^T$$

with U orthogonal and Λ diagonal. By Linear transformation we can decompose W as

$$\begin{bmatrix} U & 0 \\ 0 & U \end{bmatrix}^T D \begin{bmatrix} U & 0 \\ 0 & U \end{bmatrix}$$

with

$$D := \begin{bmatrix} D_1 & & & \\ 0 & D_2 & & \\ \dots & & & \\ 0 & 0 & \dots & D_n \end{bmatrix}$$

such that

$$D_i := \begin{bmatrix} (1+\beta)(1-\alpha\lambda_i) & -\beta(1-\alpha\lambda_i) \\ 1 & 0 \end{bmatrix}.$$

Choosing the steplength as the in the assumption we have

$$\rho(T) \leq 1 - \frac{1}{\sqrt{\kappa}} < 1$$

□

Remark.

1. For certain class of functions we observe that constant steplength may not be the best choice.
2. The optimal convergence rate: 1. Convex function: (quadratic sublinear) sublinear.
3. NAG doesn't cover the quasi-Newton method: super-linear rate possible.

Theorem 2.4. *More generally, for $f : \mathbb{R}^n \to \mathbb{R}$ that are strongly convex with the constant m and L-smooth with the constant L, the **NAG** method with the steplength*

$$\alpha_k = \frac{1}{M}$$
$$\beta_k = \frac{1-\sqrt{\kappa}}{1+\sqrt{\kappa}}$$

*converges **q-linearly** with $\rho = (1 - \frac{1}{\sqrt{\kappa}})$ and the convergence rate*

$$f(x_k) - f_* \leq (1 - 1/\sqrt{\kappa})^k \left(f(x_0) - f_* + \frac{m}{2}\|x_0 - x_*\|^2 \right). \tag{II.29}$$

Proof. Define the Lyapunov function

$$V_k = f(x_k) - f(x_*) + \frac{L}{2}\|\tilde{x}_k - \rho^2 \tilde{x}_{k-1}\|^2. \tag{II.30}$$

We show that

$$V_{k+1} \leq \rho^2 V_k.$$

□

Theorem 2.5 (super-linear convergence for BFGS method)**.** *For $f \in C^2(\mathbb{R}^n, \mathbb{R})$,*

Proof. □

2.3 Method of the conjugated gradient

What is the conjugate method?

Observing the optimality condition of the quadratic objective function is a linear equation, we may solve the linear equation by the Gramer's rule.

Till now, all the algorithm above need the Lipschitz constant of the gradient except for **NAG** for the strongly convex functions.

Observe the Heavy Ball method, for the quadratic function $f = \frac{1}{2}\langle x, Ax\rangle - \langle b, x\rangle + c$ we have

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k) + \beta_k (x_k - x_{k-1}).$$

Given $f : \mathbb{R}^n \to \mathbb{R}, f(x) = \langle Ax, Ax\rangle - \langle b, x\rangle + c$.

$x_0 \in \mathbb{R}^n$, $d_0 = \nabla f(x_0)$, $\|\nabla f(x_k)\| > \varepsilon \|f(x_k)\|$ $x_{k+1} = x_k + \alpha_k d_k$, $d_{k+1} = -\nabla f(x_k) + \mu_k d_k$, $r_{k+1} = r_k + \alpha_k A d_k$.

Write as $x_{k+1} = x_k + \alpha_k d_k$.

$$\alpha_k d_k = x_{k+1} - x_k = -\alpha_k \nabla f(x_k) + \frac{\beta_k}{\alpha_{k-1}} d_{k-1},$$
$$d_k = -\nabla f(x_k) + \underbrace{\left(\frac{\beta_k}{\alpha_k}\alpha_{k-1}\right)}_{\mu_k} d_{k-1}.$$

Define the residual $r_k = \nabla f(x_k) = Ax_k - b$ we have

$$r_{k+1} = Ax_{k+1} - b = Ax_k - b + \alpha_k A d_k = r_k + \alpha_k A d_k.$$

Now the question: **How do we choose the steplength α_k, β_k?**

1. Choose α_k as

$$\arg\min_{\alpha} f(x_k + \alpha d_k) = \frac{\langle d_k, r_k\rangle}{d_k^T A d_k}$$

2. To have the conjugation $\langle d_{k+1}, A d_k\rangle =$ we have

$$\mu_k = \frac{\|\nabla f(x_{k+1})\|^2}{\|\nabla f(x_k)\|^2} = \frac{\|r_{k+1}\|^2}{\|r_k\|^2}$$

2.4 Method of Fetcher-Rieves

3 Stochastic gradient method

3.1 Motivation

Idea:

1. Large amount of data
2. Calculation of the gradients expensive(Why?)
3. frequent structure of the target function:

$$f := \frac{1}{M}\sum_{i=1}^{M} f_i(x) \ \texttt{for } M \texttt{ large}$$

We introduce 2 approximation methods:

(mini)batch approximation:

$$\nabla f(x) \approx \frac{1}{|I_k|}\sum_{i \in I_k} \nabla f_i(x)$$

where $I_k \subset M$ randomly selected index set. Or in the extreme case:

stochastic gradient method:

$$\nabla f(x) \approx \nabla f_i(x) \ \texttt{where } i \texttt{ randomly selected}$$

For the Stochastic Gradient Method we need:

1. Generator of a random variable ξ_k with $k \in \mathbb{N}$.

2. Given ξ_k and x_k we calculate

$$g(x_k, \xi_k) = \nabla f_{\xi_k}(x_k)$$

3. The step size α_k.

Theorem 3.1 (Theorem of Rademacher)**.** *The finite dimensional Lipschitz continuous function is almost a.e. differentiable.*

3.2 Analysis of the SGM

Remark. Observe that

1. No termination criteria is specified(may not converge to the minimum).

2. On the choice of α_k: 1. constant(frequently used in machine learning) 2. $\alpha_k \to 0$ (motivated by the sub-gradient method).

3. On the calculation of g: 1. gradient method. 2. mini-patch. 3. (Quasi-)Newton.

Assumption 3.1 (Lipschitz differentiability)**.** The objective functions $f : \mathbb{R}^n \to \mathbb{R}$ is L-smooth.

The inequality of L-smoothness (II.7) is no longer **deterministic** in the case of stochastic gradient method. Under the assumption 3.1 we get:

Lemma 3.2. *Under the Assumption 3.1, the SGM generates a sequence s.t.*

$$\mathbb{E}_{\xi_k} f(x_{k+1}) - f(x_k) \leq -\alpha_k \mathbb{E} g(x_k, \xi_k) \nabla f(x_k) + \frac{L}{2} \alpha_k^2 \mathbb{E} g(x_k, \xi_k)^2 \tag{II.31}$$

Proof. Taking the expectation of the (II.7). □

Assumption 3.2. For the function $f : \mathbb{R}^n \to \mathbb{R}$ we have

1. x_k generated by Algorithm 3.2 stays in an open set. And f is bounded from below.

2. there exits $\mu_g \geq \mu > 0$ s.t.

$$\mu \|\nabla f(x_k)\| \leq \mathbb{E}_{\xi_k} \|g(x_k, \xi_k)\| \leq \mu_g \|\nabla f(x_k)\| \tag{II.32}$$

3. there exits $M_g \geq M > 0$ s.t.

$$\mathtt{Var}[g(x_k, \xi_k)] \leq M_g \|\nabla f(x_k)\|^2 + M_R \tag{II.33}$$

Lemma 3.3. *Under the assumption 3.1 and assumption 3.2 we have*

$$\mathbb{E}_{\xi_k}[f(x_{k+1})] - f(x_k) \leq (-\mu + \frac{1}{2}\alpha_k L M_g)\alpha_k \|\nabla f(x_k)\|^2 + \frac{1}{2}\alpha_k^2 L M_R \tag{II.34}$$

Theorem 3.4 (Convergence Rate of strongly convex functions)**.** *Given assumption 3.1 and assumption 3.2 and that f is strongly convex. Choose*

$$\alpha_k = \alpha \leq \frac{\mu}{L M_g}. \tag{II.35}$$

we have the r-linear convergence rate:

$$\mathbb{E}[f(x_k)] - f_* \leq \tilde{C} + (1 - \alpha c \mu)^k \left(f(x_0) - f_* - \tilde{C} \right) \tag{II.36}$$

with $\tilde{C} = \frac{\alpha L M_R}{2c\mu}$, i.e., the algorithm converges against

$$\tilde{C} = \frac{\alpha L M_R}{2c\mu}.$$

Proof. Under the assumption we have

$$-\mu + \frac{1}{2}\alpha_k L M_g \leq -\frac{\mu}{2} < 0$$

By the last lemma and (II.15) we have

$$\begin{aligned}
\mathbb{E}[f(x_k)] - f(x_k) &\leq -\frac{\mu}{2}\alpha\|\nabla f(x_k)\|^2 + \frac{L}{2}\alpha^2 M_R \\
&\leq -\mu c\alpha(f(x_k) - f_*) + \frac{L}{2}\alpha^2 M_R \\
\mathbb{E}[f(x_{k+1})] - f_* &\leq (1 - \alpha\mu c)\Big(f(x_k) - f_*\Big) + \frac{L}{2}\alpha^2 M_R.
\end{aligned}$$

Taking expectation over all ξ_i we get

$$\mathbb{E}[f(x_k) - f_*] \leq (1 - \alpha c\mu)\mathbb{E}[f(x_k) - f_*] + \frac{L}{2}\alpha^2 M_R. \tag{II.37}$$

rearrange the term we get (II.36). The contraction follows from:

$$\alpha c\mu \leq \frac{\mu^2 c}{L M_g} \leq \frac{c}{L} < 1.$$

□

Remark.

1. If $g(\xi_k, x_k)$ is the unbiased estimator of $\nabla f(x_k)$ we have $\mu = \mu_g = 1$. If it doesn't have the noise we have $M_R = 0, M_g = 1$. In this case we have the constant step size $\alpha = \frac{1}{L}$ and we goes back to the steepest descent method with constant steplength.

2. The Termination criteria: The convergence rate shows that in the algorithm $\mathbb{E}[f(x_k)] - f_*$ converges to the constant $\frac{\alpha L M_R}{2c\mu}$. We may reduce the steplength α each time by half and have the $\mathbb{E}[f(x_k)] - f_*$ converges to 0. However, it raises the problem that f_* is not known.

Remark.

1. The convergence rate is sublinear.

2. The termination criteria is not applicable since f_* is unknown.

The variable steplength.

By (II.36) we observe the tradeoff between convergence speed and the precision of the algorithm. To achieve the ideal balance we conduct the following scheme:

Termination criteria: $\mathbb{E}[f(x_k) - f_*] \leq 2\tilde{C}$. Reduce the step size then by half. We obtain 2 sequence

$\{\alpha_i\}$: `sequence of step size`

$\{k_i\}$: `sequence of iteration`

$\{f_{\alpha_i}\}$: `sequence of precision`

Theorem 3.5 (Robbins and Monro)**.** *Given assumption 3.1 and 3.2, and f strongly convex with $c > 0$. Define for the stochastic gradient method 3.2 the step size*

$$\begin{aligned}
&\alpha_1 \leq \frac{\mu}{L M_g} \\
&\alpha_k = \frac{\beta}{\gamma + k} \quad \textit{with } \beta > \frac{1}{c\mu}, \gamma > 0
\end{aligned} \tag{II.38}$$

then we have

$$\mathbb{E}[f(x_k) - f_*] \leq \frac{\nu}{\gamma + k} \tag{II.39}$$

with $\nu(x_0, \beta) = \max\left\{\frac{\beta^2 L M_R}{2(\beta c\mu - 1)}, (\gamma + 1)(f(x_0) - f_*)\right\}$.

Proof. We have by the last theorem

$$\mathbb{E}[f(x_{k+1}) - f_*] \leq \left(1 - \frac{\beta cu}{\gamma + k}\right)\frac{\nu}{\gamma + k} + \frac{L}{2}\left(\frac{\beta}{\gamma + k}\right)^2 M_R \tag{II.40}$$

Induction with (II.36). □

Remark.

1. **Choice of** x_0:
2. **Choice of** β: $\beta = \frac{2}{c\mu}$

Theorem 3.6. *Assuming assumption 3.1 and 3.2. Choose step size* $\alpha < \frac{\mu}{LM_g}$. *The generated sequence satisfies*

$$\mathbb{E}[\frac{1}{K}\sum_{k=0}^{K-1}\|\nabla f(x_k)\|^2] \leq \frac{\alpha L M_R}{\mu} + \frac{2}{\mu\alpha K}\Big(f(x_k) - f(x_*)\Big) \tag{II.41}$$

Remark.

1. In the case of null quadratic variation, the algorithm is deterministic and $\|\nabla f(x_k)\| \to 0$.
2. In the case of positive quadratic variation, no statement regarding the optimal distance to the minimum (since the gradient is never 0 and the necessary optimality condition is not satisfied).
3. But with vanishing step size we show the convergence.

Theorem 3.7. *Under the assumption 3.1 and 3.2, if the step size satisfies*

$$\sum \alpha_k = \infty, \quad \sum \alpha_k^2 < \infty \tag{II.42}$$

then define the partial sum

$$A_k := \sum_{k=0}^{K-1} \alpha_k$$

we have

$$\lim_k \mathbb{E}\left[\frac{1}{A_k}\sum_{k=0}^{K-1}\alpha_k\|\nabla f\|^2\right] \to 0 \tag{II.43}$$

Proof. Apply the assumption on the (II.34) in the lemma above and get the result. □

Theorem 3.8. *Under assumption 3.1 and 3.2, if the step size satisfies*

$$\sum \alpha_k = \infty, \quad \sum \alpha_k^2 < \infty \tag{II.44}$$

then the sequence x_k *generated by the stochastic gradient method 3.2 satisfies*

$$\liminf_k \mathbb{E}\|f(x_k)\|^2 \to 0 \tag{II.45}$$

Proof. □

The Mini-Batch method

Back to the mini-batch method

$$g(x_k, I_k) = \frac{1}{|I_k|} \sum_{\xi_i \in I_k} g(x_k, \xi_k). \tag{II.46}$$

Define $N := |I_k|$, ny the variance sum law we have

$$\operatorname{Var} g(x_k, \xi_k) = N \operatorname{Var} g(x_k, I_k),$$

i.e., the mini-batch reduced the variance. By (II.36) we get the optimal distance $\tilde{C} = \frac{\alpha L M_R}{2c\mu}$ is reduced by a factor of N with the same step size.

Recall

$$\alpha_0 < \frac{\mu}{L M_g}.$$

If we reduce the M_g by N, it's not clear whether we can choose α_0 big enough.

The computational complexity in the stochastic setting

Convergence proofs have to be adapted since we only have a.s.bounded gradient. There are examples, where the optimisation method of the smooth function fails in the non-smooth case.

Alternative of Computational Complexity and Termination.

Given scenarios:
1. huge number of data points M
2. Amount of computing time $\mathcal{T}$.
Introduce the target function:

$$f_e(x) := \mathbb{E}_{w,y}[l(w, y, x)] \tag{II.47}$$

1. $x_* \in \arg\min_x f_e(x)$: `the true estimation`
2. $x_M \in \arg\min_x f(x)$: `the empirical estimation`
3. $\tilde{x}_M$: `estimation by limited time` $\mathcal{T}$.

The objective function becomes

$$\min \mathbb{E}_x f_e(\tilde{x}_M)$$

We evaluate the target by:

$$\mathbb{E} f_e(\tilde{x}_M) = \underbrace{f_e(x_*)}_{(1)\ \texttt{a priori error(by model)}} + \underbrace{\mathbb{E}[f_e(x_M) - f_e(x^*)]}_{(2)\texttt{sampling error(by sample size)}} + \underbrace{\mathbb{E}[f_e(\tilde{x}_M) - f_e(x_M)]}_{(3)\texttt{time restriction error(by } \mathcal{T}\texttt{)}}$$

By increasing M:
1. reduced approximation error
2. increased time restriction error ε.
We denote
1. $\mathcal{T}(M.\varepsilon)$: Time needed by the optimisation approach to achieve time **restriction error** ε.

Now we concentrate on the optimisation problem:

$$\min \mathcal{E}(M,\varepsilon) := \mathbb{E}[f_e(\tilde{x}_M) - f_e(x^*)]$$

`s.t.`

$$\mathcal{T}(M,\varepsilon) \leq \mathcal{T}$$

1). For the SGM we can derive:

$$\mathcal{E}(M,\varepsilon) = \mathcal{O}\left(\sqrt{\frac{\log(M)}{M} + \varepsilon}\right)$$

2). If we assume additional assumption on the cost function(strong convexity), it can be improved to

$$\mathcal{E}(M,\varepsilon) = \mathcal{O}\left(\frac{\log(M)}{M} + \varepsilon\right)$$

3). The best achievable convergence rate:

$$\mathcal{E}(M,\varepsilon) \sim \frac{1}{M} + \varepsilon$$

Furthermore assuming the minimiser of $\mathcal{E}(M,\varepsilon)$ satisfies

$$\mathcal{T}(M,\varepsilon) = \mathcal{T}$$

Then one has:

1. A SGM can achieve the ε**-optimality**: the computing time $\mathcal{T}(M,\varepsilon) \sim \frac{1}{\varepsilon}$, which is not dependent on M!
2. The (mini)batch approach can achieve ε-optimality with

$$\mathcal{T}(M,\varepsilon) \sim M\log(\frac{1}{\varepsilon})$$

3.3 Adam method and the related approaches

So far: step size either constant or piecewise constant with the property:

$$\sum \alpha_k^2 < \infty, \quad \sum \alpha_k = \infty$$

The classical nonlinear optimisation (line search) to adapt α_k to the current situation. Is there a cheap way to incorporate the local information? First approach:

$$x_{k+1,i} = x_{k,i} - \alpha_k g(x_k, \xi_k)_i \tag{II.48}$$

It may happens the one step size is not efficient for all the components, therefore:
AdaGrad proposes a componentwise upsate:

$$x_{k+1,i} = x_{k,i} - \frac{\alpha}{\sqrt{G_{k,ii} + \varepsilon}} g(x_k, \xi_k)_i \tag{II.49}$$

where G_k is defined as diagonal semi-positive definite matrix and $\varepsilon > 0$ ensures the single-step method well-defined.
Observation: usually many iteration required and the sum of the (step size)^2 may not diverge.

RMSProp:(Root Mean Square Propagation) very similar to **AdaGrad**(an adaptive learning rate method), where we consider instead of the naive sum the convex combination of the average gradient components seen so far and the current value one obtains.

$$\begin{aligned} E(g^2)_k &:= \gamma E(g^2)_{k-1} + (1-\gamma) g(x_k, \xi_k)^2 \\ \Delta x_{k,1} &:= -\frac{\alpha}{\sqrt{E(g^2)_{k,i} + \varepsilon}} g(x_k, \xi_k) \end{aligned} \tag{II.50}$$

One caveat is that in the updates the units do **NOT** fit.

Adam(the adaptive moment estimation 2015.)The step size is based on the estimation of the 1. and the 2. momentum.

- relation the the momentum method
- can be seen as scaling of SGM with diagonal matrix.

Remark.

1. $\alpha = 0.001, \beta_1 = 0.9, \beta_2 = 0.999, \varepsilon = 10^{-8}$.

2. The initialisation of the momentum causes a bias towards 0. The rescaling tries to reduce the bias.

$$\mathbb{E}\hat{v}_k = \mathbb{E}g_{k-1}^2(1-\beta_2^k) + \xi \qquad \texttt{for } \xi \texttt{ small}$$

3. The direction obtained by Adam is scaling invariant. Scaling mean linear transformation by a diagonal matrix: $cg(x_k, \xi_k)$.

The author proposed the method together with a convergence result: Consider

$$R(k) := \sum_{k=1}^{K} f_k(x_k) - \min_{x \in X} \sum_{k=1}^{K} f_k(x)$$

$$\frac{R(K)}{K} = \mathcal{O}(\frac{1}{k})$$

In practice, **Adam** works very well, it is THE method for general problem

Notataion:

$S^+ =$ set of all positive definite matrices in $\mathbb{R}^{k \times k}$

$X \subset \mathbb{R}^n$ = feasible set.

This generic algorithm covers all the stochastic methods introduced so far:

SGM:

$$g_k = g_{\xi_k}$$
$$\phi_k(g_0, \ldots, g_k) = g_k$$
$$\psi_k(g_0, \ldots, g_k) = I$$

AdaGrad:

$$g_k = g_{\xi_k}$$
$$\phi(g_0, \ldots, g_k) = g_k$$
$$\psi_k(g_0, \ldots, g_k) = \texttt{diag}\{g_0^2 \ldots, g_k^2\}/(k+1)$$

RMSprop

$$g_k = f_{\xi_k}$$
$$\Phi_k(g_0, \ldots, g_k) = g_k$$
$$\Psi_k(g_0, \ldots, g_k) = (1-\beta_2)g_k^2 + \beta_2 v_k$$

Adam

$$g_k = f_{\xi_k}$$
$$\Phi_k = (1-\beta_1)\sum_{i=1}^{k} \beta_1^{k-i} g_{i-1}$$
$$\Psi_k = (1-\beta_2)\sum_{i=1}^{k} \beta_2^{k-i} g_{i-1}^2$$

Now define

$$R(k) = \sum_{i=0}^{k} f_k(x_k) - \min_{x \in X} \sum_{i=0}^{k} f_k(x) \tag{II.51}$$

Assumptions:

1. $\|\nabla f\| < M$ `on` X

2. X `bounded`

Goal: $R(k) = o(k)$ (Convergence not only in the expectation.)

Theorem 3.9. *There exists **convex** optimisation problems, s.t., the Adam method doesn't converge, i.e.,*

$$\frac{R(k)}{k} \nrightarrow 0 \quad \texttt{as} \quad k \to \infty \tag{II.52}$$

Theorem 3.10. *For $\beta_1 \geq \sqrt{\beta_2}$, there exists optimisation problem such that*

$$\frac{R(k)}{k} \nrightarrow 0 \quad \texttt{as} \quad k \to \infty$$

4 LASSO problems

LASSO stands for **Least Absolute Selection and Shrinking Operation** (1996): Once more: Description of relation between inputs and outputs

$$\texttt{before:} \quad (w,v) \mapsto \int l(w,v)dP(\Omega) = \frac{1}{M}\sum_{i=1}^{M} f_i(x)$$

`here: different model, usually linear relation`

$x \in \mathbb{R}^p$ `set of parameter`, $y \in \mathbb{R}^m$ `output`

`s.t.`

$y = Ax, \quad A \in \mathbb{R}^{m \times p}$

Usually the equality constraints can not be fulfilled, we therefore consider the least square problem:

$$\min_{x \in \mathbb{R}^p} \|y - Ax\|^2$$

There are methods, such as QR decomposition, to solve this kind of problem.

4.1 Analysis of the problem setting

From the optimisation point of view:

$$\tilde{f}(\tilde{y}) = \|y - \tilde{y}\|^2 \quad (\tilde{y} = Ax)$$

Minimisation w.r.t. $\tilde{y}$ yield a strongly convex optimisation problem. $\Rightarrow$ solution is unique.
Impact of A (with $\tilde{y} = Ax$) on the existence and uniqueness of the minimiser x?
Case 1:

- $m = p$
- A quadratic and invertible. (x `such that` $Ax = \tilde{y}$ is unique)

Case2:

- $m < p$ (underdetermined system) & $\text{ran}(A) = m$ (A has full rank)
- solution set $\mathcal{L} = \{x \in \mathbb{R}^p | Ax = y\} = \{x = x_+ + h, h \in \ker(A)\}$ a $p - m$ dimensional minifold.
- Task:
  - Find x_+ and basis of $\ker(A)$.
  - search for x_+ with minimal norm, or even a **sparse** solution.

Case 3:

- $m > p$ (overdetermined system)
- A solution exists if $y \in \texttt{Im}(A)$, in general no exact solution exists
- Task

$$\min_x \|Ax - y\| \ \texttt{x is the best approximtiobn}$$

i.e.,

$$\min f(x) = \frac{1}{2}\|Ax - y\|^2$$
$$\texttt{s.t.}$$
$$\nabla f(x) = 0 \Leftrightarrow A^T Ax = A^T y \Leftrightarrow x = (A^T A)^+ A^T y \quad \textbf{(NOC)}$$

The problem is that $\dim A$ very large and normal equations has large condition numbers (sensitive to the numerical error). An alternative is needed!

LASSO(the least absolute selection shrinkage operator)

1. One wants to solve

$$\min_{x \in \mathbb{R}^p} \|y - Ax\|_2^2$$
$$\texttt{s.t.}$$
$$\|x\|_0 \leq k$$

Caveat: the $\|\cdot\|_0$ is **not** a norm and this problem is **NP-hard**.
2. Therefore: Ridge regression

$$\min_{x \in \mathbb{R}^p} \|Ax - y\|_2^2$$
$$\texttt{s.t.}$$
$$\|x\|_2^2 \leq k$$

Problem: solution is usually not sparse
3. Further: **Constrained LASSO**

$$\begin{aligned} &\min_{x \in \mathbb{R}^p} \|Ax - y\|_2^2 \\ &\texttt{s.t.} \\ &\|x\|_1 \leq k \end{aligned} \tag{II.53}$$

4. Geometrically showed it approximates the sparse solution. One can show it's equivalent to **Unconstrained LASSO:**

$$\begin{aligned} &\min \|y - Ax\|_2^2 + \lambda \|x\|_1 \\ &\texttt{s.t.} \\ &x \in \mathbb{R}^p \end{aligned} \tag{II.54}$$

for a suitable choice of λ, and hence transform the constrained optimisation problem to a unconstrained optimisation problem.

Theorem 4.1. *The constraint LASSO problem is equivalent to an unconstrained LASSO problem of the Lagrange function.*

Proof.
1. Existence of solution of the constrained optimisation problem provided by the Cantor theorem, denoted by x_*.

$$\begin{aligned} &\min_{x \in \mathbb{R}^p} \|Ax - y\|^2 \\ &\texttt{s.t.} \\ &g(x) \leq 0 \end{aligned}$$

2. The existence of the minimiser of the unconstrained LASSO is provided by the duality theory and Lagrange function, i.e., define the Lagrange function:

$$L(x, \lambda) := f(x) + \lambda^T g(x) \ \texttt{with} \ \lambda \in \mathbb{R}_+^m, \quad g(x) \leq 0,$$

for

$$M := \{x \in \mathbb{R}^m : g(x) \leq 0\},$$

take the supremum over all the positive Lagrange multipliers:

$$\Phi(x) := \sup_{\lambda \in \mathbb{R}_+^m} L(x, \lambda),$$

we have

$$\Phi(x) = \begin{cases} f(x) & x \in M, \\ \infty & x \notin M. \end{cases}$$

We get an equivalent formulation of the minimisation problem

$$\inf_{x \in \mathbb{R}^n} \Phi(x) = \inf_{x \in M} f(x) = f(x_*) := F(x). \tag{II.55}$$

switch the inf and sup we yield the dual problem:

$$\sup_{\lambda} G(\lambda), \qquad G(\lambda) = \inf_{x \in \mathbb{R}^p} L(x, \lambda) \tag{II.56}$$

Then we have for $x \in M$

$$G(\lambda) = \inf_{x \in M} \Big(f(x) + \lambda g(x) \Big) \leq f(x_*) = F(x). \tag{II.57}$$

The condition is that the **weak duality**. And the **linear constraint and quadratic objective function** provides the **strong duality**(Slater condition):

$$G(\lambda) = F(x). \tag{II.58}$$

□

In the literature one finds several formulations of the LASSO problem.

$$\min \frac{1}{2}\|Ax-y\|^2+\lambda_1\|x\|_1$$
$$\min \frac{1}{m}\|Ax-y\|^2+\frac{1}{m}\lambda_2\|x\|_1$$
$$\min \frac{1}{2m}\|Ax-y\|^2+\frac{1}{2m}\lambda_2\|x\|_1$$

Here we take the first formulation.

4.2 Properties of the solution of the LASSO problem

Target function

$$f(x)=\frac{1}{2}\|Ax-y\|^2+\lambda\|x\|_1=\tilde{f}(x)+r(x)$$

Definition 4.1 (coercivity)**.** Coercivity implies infinite growth at infinity, i.e., if for all $\{x\}_k \subset \mathbb{R}^p$ we have

$$\|x_k\| \to \infty \Rightarrow \|f(x_k)\| \to \infty \tag{II.59}$$

the function is coercive (for any starting point).

Theorem 4.2. *Let $f:\mathbb{R}^n \to \mathbb{R}$ be continuous and coercive, then one has that the problem $\min_{x\in\mathbb{R}^n} f(x)$ has at least one solution*

Proof. Follows from the standard methods form the variational calculus. □

By coercivity we know there exists at least one solution, by convexity we know the convex combination of solutions are also solutions, which also motivates to find sparse solutions.

Generalised NOC

Since the L^1 norm is **not differentiable** we need to use the subdifferential instead of the differential. We derive the **NOC** with x_* the minimiser and s the subderivative.

$$A^T(Ax_*-y)=\lambda s \in \mathbb{R}^p$$

with s defined as

$$s \in \begin{cases} \texttt{sign}(x_{*,i}) & x_{*,i} \neq 0 \\ [-1,1] & x_{*,i}=0 \end{cases} \tag{II.60}$$

Since the LASSO problem is convex, we can see the necessary condition is also **sufficient.**

Interesting properties:

1. Componentwise, all solutions have the same sign: for x_1^*, x_2^* 2 solutions,

$$!\nexists j \in \texttt{supp}(x_*^1)\cap\texttt{supp}(x_*^2) : \texttt{sign}(x_*^1) \neq \texttt{sign}(x_*^2)$$

2. Each solution yields the same:

$$\tilde{y}=Ax$$

3. If $\lambda > 0$ each solution has the same L^1 norm

Proof.
2). Assume the contradiction, $\|Ax_1\|_1 \neq \|Ax_2\|_1$, by the strict convexity of the quadratic function we have

$$\begin{aligned} f_* &= \min_{x\in\mathbb{R}^p} f(x) \\ &= \frac{1}{2}\|A(\alpha x_1 + (1-\alpha)x_2)\|_2^2 + \lambda\|\alpha x_1 + (1-\alpha)x_2\|_1 \\ &< \alpha f(x_1) + (1-\alpha) f(x_2) \\ &= f_* \end{aligned}$$

□

1. Dependence on λ: One can show that $x_*(\lambda)$ is a piecewise linear function. (for Ridge regression the smooth dependence). By choosing large λ the $\|x\|_1$ tends to 0.
2. $\|\cdot\|_1$ not differentiable: In general it is not possible to find a solution of the unconstrained optimisation problem directly, corresponding algorithm:

- **ADMM**: Alternative direction method of multipliers
- **ISTA**: Iterative shrinkage threshold algorithm

4.3 ADMM Method

Again motivated by duality we consider

$$\begin{aligned} &\min_{(x,z)\in\mathbb{R}^{p+1}} f(x) + h(z) \\ &\texttt{s.t.} \\ &Ax + Bz = c \end{aligned} \tag{II.61}$$

with $A \in \mathbb{R}^{m\times p}, B \in \mathbb{R}^{m\times n}, c \in \mathbb{R}^m$ and f, h are convex function.
The corresponding Lagrange function reads

$$L(x,z,\lambda) = f(x) + h(z) + \lambda(Ax + Bz - c)$$

The dual problem is given by

$$\begin{aligned} &\min_{\lambda\in\mathbb{R}^m_+} q(\lambda) \\ &\texttt{with} \\ &q(\lambda) = \min_{x,z} L(x,z,\lambda) \end{aligned}$$

If the strong duality holds (the dual problem and the final problem are equivalent), it follows for the solution (x_*, z_*, λ_*):

$$f(x_*) + h(z_*) = q(\lambda_*)$$

One approach to solve the dual problem is

naive ansatz:

For the primal step we consider

$$L(x,z,\lambda) = \underbrace{f(x) + \lambda^T(Ax - \frac{1}{2}c)}_{:=L_1} + \underbrace{h(z) + \lambda^T(Bz - \frac{1}{2}c)}_{:=L_2}$$

compute:

$$x_{k+1} \in \arg\min L_1(x,\lambda_k)$$
$$z_{k+1} \in \arg\min L_2(z,\lambda_k)$$

We have 2 smaller subproblem.

ADMM ansatz

For ADMM one exploits that for $\rho > 0$ the unconstrained problem is equivalent to

$$\min_{x,z} f(x) + h(z) + \frac{\rho}{2}\|Ax + Bz - c\|^2$$
$$\texttt{s.t.}$$
$$Ax + Bz = c$$

If A, B chosen appropriately. The **augmented** Lagrange function of the extended problem is given by:

$$L_\rho(x,z,\lambda) = f(x) + h(z) + \lambda^T(Ax + Bz - c) + \frac{\rho}{2}\|Ax + Bz - c\|^2 \tag{II.62}$$

Define the residual: $r = Ax + Bz - c$, we may rewrite the **augmented Lagrangian:**

$$L_\rho(x,z,\lambda) = f(x) + h(z) + \frac{\rho}{2}\left(\|r + \frac{1}{\rho}\lambda\|^2 + c(\lambda)\right). \tag{II.63}$$

Assumption 4.1. The augmented Lagrange function L_ρ has a saddle point for $\rho = 0$, i.e., there exists (x_*, z_*, λ_*) s.t.

$$L_0(x_*, z_*, \lambda) \le L_0(x_*, z_*, \lambda_*) \le L_0(x, z, \lambda_*) \quad \forall x, y, z.$$

This ensures the strong duality.

Theorem 4.3. *Let the assumption 4.1 holds, then one has*

1. *the residual converges*

$$\lim_k r_k \to 0.$$

2. *the value of the target function converges, i.e.,*

$$\lim_k f(x_k) + h(z_k) \to \min_k \left(f(x_k) + h(z_k)\right). \tag{II.64}$$

3. *It holds*

$$\lim_k \rho \langle A, B(z_{k+1} - z_k)\rangle = 0. \tag{II.65}$$

Proof. see literature. □

Remark.

1. No statement regarding the convergence of x_k, z_k and the Lagrange multipliers λ_k. For that one need additional properties. (like before, we have the convergence of the gradient but not convergence of the iterates in the stochastic gradient method).

2. The primal admissibility: by $\lim_k r_k \to 0$ we know the primal admissibility: $Ax + Bz - c = 0$ is satisfied at the limit.

3. The dual admissibility: The dual admissibility is defined as

$$\nabla_{x,z} L_\rho = 0. \tag{II.66}$$

By $\lim_k \rho \langle A, B(z_k - z_{k+1})\rangle \to 0$ we know the

Since f and h are assumed to be convex, the optimality conditions are necessary and sufficient.

From ADMM to LASSO

For the LASSO problem we apply

$$f(x) = \frac{1}{2}\langle Ax - b, Ax - b\rangle$$

$$h(z) = \lambda \|z\|_1 \ \texttt{λ not Lagrange multiplier}$$

$$x - z = 0 \ \texttt{the equality constraints}(A = I, B = -I, c = 0)$$

Then the algorithm w.r.t. x can be solved explicitly, first we define the **proximal operator:**

$$\texttt{prox}_h(x) = \arg\min_u (h(u) + \frac{1}{2}\|u - x\|_2^2) \tag{II.67}$$

Lemma 4.4 (update of x)**.** *For the general quadratic function f with Q SPD:*

$$f(x) = \frac{1}{2}\langle Q, Qx\rangle + \langle b, x\rangle + c$$

The primal step for x can be calculated as

$$\texttt{prox}_f(y) = (Q + I)^{-1}(y - b)$$

and for the LASSO problem we have

$$x_{k+1} = \texttt{prox}_{\frac{f}{\rho}}(z_k - \frac{1}{\rho}\lambda_k) = (A^T A + \rho I)^{-1}(\rho z_k - \lambda_k + A^T b) \tag{II.68}$$

Proof. The quadratic function $\frac{1}{2}\|Ax - b\|^2$ can be written in the canonical form

$$f(x) = \frac{1}{2}\langle x, A^T A x\rangle - \langle A^T b, x\rangle + \frac{1}{2} b^T b. \tag{II.69}$$

□

For the iteration with respect to the non-smooth L^1 norm with respect to z_k, we introduce the lemma:

Lemma 4.5 (update of z)**.** *$h : \mathbb{R}^n \to \mathbb{R}, h(u) = \lambda \|u\|_1$ with $\lambda > 0$, it follows that*

$$\texttt{prox}_h(x) = \mathcal{S}_\lambda(x),$$

$$\texttt{where}$$

$$S_\lambda(x_i) = \lfloor |x| - \lambda I_n \rfloor \cdot \texttt{sign}(x).$$

and for the update of z we have

$$z_{k+1} \in \texttt{prox}_{\frac{h}{\rho}}(x_{k+1} + \frac{1}{\rho} y_k) = \mathcal{S}_{\frac{\lambda}{\rho}}(x_{k+1} + \frac{1}{\rho}\lambda_k) \tag{II.70}$$

Given $f : \mathbb{R}^n \to \mathbb{R}$

$x_0 \in \mathbb{R}^n$ k = 0,1, 2... `generate` ξ_k `calculate` $g(x_k, \xi_k)$ `choose` $\alpha_k = \alpha < \frac{\mu}{M_g L}$ $x_{k+1} = x_k - \alpha_k g(x_k, \xi_k)$

$\alpha_0 = \frac{\mu}{LM_g}, f_{\alpha_0} = \alpha_0$ $f_{\alpha_i} \geq \varepsilon$ $f(x_k) - f_* \leq f_{\alpha_i}$ `generate` ξ_k $x_{k+1} = x_k - \alpha_i g(x_k)$. $\alpha_{i+1} = \frac{\alpha_i}{2}$, $f_{\alpha_{i+1}} = \frac{f_{\alpha_i}}{2}$.

Given $f : \mathbb{R} \to \mathbb{R}, x_0 \in \mathbb{R}^n, \alpha > 0, \beta_1, \beta_2 \in [0,1]$

$m_0 = 0$, $v_0 = 0$ $\mathcal{T}$ generate ξ_k $m_k = \beta_1 m_{k-1} + (1-\beta_1) g(x_k, \xi_k)$ $v_k = \beta_2 v_{k-1} + (1-\beta_2) g(x_k, \xi_k)^2$ $\hat{m}_k = \frac{m_k}{1-\beta_1^k}, \hat{v}_k = \frac{v_k}{1-\beta_2^k}$ (bias correction) $\hat{x}_k = x_k - \alpha \frac{\hat{m}_k}{\sqrt{\hat{v}_k}}$

Given $x_0 \in \mathbb{R}^n, f_k : \mathbb{R}^n \to \mathbb{R}, \alpha_k > 0$, function

$\phi_k : \mathbb{R}^{(k+1)n} \to \mathbb{R}^n$ `1st momentum generator`

$\psi_k : \mathbb{R}^{(k+1)n} \to S^+$ `2nd momentum generator`

$x_0 = 0, m_0 = 0, v_0 = 0$ $\mathcal{T}$ $m_k = \Phi_k(g_0, \ldots, g_k)$ $v_k = \Psi_k(g_0, \ldots, g_k)$ $\hat{x}_k = x_{k-1} - \alpha_k \frac{m_k}{\sqrt{v_k^2}}$ $x_k = \mathrm{P}_X \hat{x}_k$

Given $f : \mathbb{R}^p \to \mathbb{R}, h : \mathbb{R}^n \to \mathbb{R}$.

$\lambda_0 \in \mathbb{R}^m, x_0 \in \mathbb{R}^p, z_0 \in \mathbb{R}^m$. $\mathcal{T}$ $(x_{k+1}, z_{k+1}) \in \arg\min_{x,z} L(x, y, \lambda_k)$ `(the primal step)`, $\lambda_{k+1} = \lambda_k + \alpha_k \nabla_\lambda L(x_{k+1}, z_{k+1}, \lambda_k)$ `(dual step)`.

Given $f : \mathbb{R}^p \to \mathbb{R}, h : \mathbb{R}^n \to \mathbb{R}$

$z_0 \in \mathbb{R}^n, \lambda_0 \in \mathbb{R}^m, \rho > 0$ $\mathcal{T}$ $x_{k+1} \in \arg\min_x L_\rho(x, z_k, \lambda_k)$ `(updating of` x`)` $z_{k+1} \in \arg\min_z L_\rho(x_{k+1}, z, \lambda_k)$ `(updating of` z`)` $\lambda_{k+1} = \lambda_k + \rho(Ax_{k+1} + Bz_{k+1} - c)$ `(dual ascent)`

Given $f(x) = \|Ax - b\|^2, h(z) = \lambda \|z\|_1$ with $A \in \mathbb{R}^{p \times n}, b \in \mathbb{R}^p$

$z_0 \in \mathbb{R}^n, \lambda_0 \in \mathbb{R}^n$ $k = 0, 1, \ldots$ $x_{k+1} = \mathrm{prox}_{\frac{f}{\rho}}(z_k - \frac{1}{\rho}\lambda_k) = (A^T A + \rho I)^{-1}(\rho z_k - \lambda_k + A^T b)$, $z_{k+1} = \mathcal{S}_{\frac{h}{\rho}}(x_{k+1} + \frac{1}{\rho}\lambda_k)$, $\lambda_{k+1} = \lambda_k + \rho(x_{k+1} - z_{k+1})$.

Proof. trivial. □

Remark.

1. Every step can be calculated explicitly
2. By constant step size $\rho \equiv const$ we have the constant $(A^T A + \rho I)^{-1}$. The cost is largely reduced since inversion for $A \in \mathbb{R}^{n\times n}$ costs (n^3).
3. **Convergence rate of Augmented Lagrangian Method:** super-linear for variable ρ.
4. **Convergence rate of ADMM:** Best case linear rate.

4.4 ISTA

Summary: solve LASSO using proximal approximation:
Observe for $f \in C^1(\mathbb{R}^n)$ the optimisation problem

$$\begin{aligned}
&\min_x f(x),\\
&\texttt{s.t.}\\
&x \in \mathbb{R}^n
\end{aligned}$$

the gradient descendent method provides the iteration:

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k),$$

which is equivalent to

$$x_{k+1} \in \arg\min_x \left(f(x_k) + \langle x - x_k, \nabla f(x_k)\rangle + \frac{1}{2\alpha_k}\|x - x_k\|^2 \right)$$

Now observe again for $f \in C^1(\mathbb{R}^n)$

$$\begin{aligned}
&\min f(x) + \lambda \|x\|_1\\
&\texttt{s.t.}\\
&x \in \mathbb{R}^n
\end{aligned}$$

the iteration becomes

$$x_{k+1} \in \arg\min \left(f(x_k) + \langle x - x_k, \nabla f(x_k)\rangle + \frac{1}{2\alpha_k}\|x - x_k\|^2 + \lambda \|x\|_1 \right)$$

omit the constant term we have

$$x_{k+1} \in \arg\min \left(\|x - (x_k - \alpha_k \nabla f(x_k))\|^2 + \lambda \alpha_k \|x - x_k\|_1 \right) = \mathcal{S}_{\lambda\alpha_k}\left(x_k - \alpha_k \nabla f(x_k) \right), \tag{II.71}$$

and for LASSO we have

$$x_{k+1} = \mathcal{S}_{\lambda\alpha_k}\left(x_k - 2\alpha_k A^T (Ax - b) \right). \tag{II.72}$$

From this we get the following

The ISTA method can be generalised to other problems

$$\begin{aligned}
&\min f(x) + g(x)\\
&\texttt{s.t.}\\
&x \in \mathbb{R}^n
\end{aligned} \tag{II.73}$$

observe now

$$\min f(y) + g(x)$$

By fixing y we observe the minimisation problem

$$Q_{\alpha,y}(x) = f(y) + \langle \nabla f(y), x - y \rangle + \frac{1}{2\alpha}\|x - y\|^2 + g(x),$$

eliminating the constant term and get the minimum

$$\begin{aligned} x_{k+1} &\in \arg\min \left(g(x) + \frac{1}{2\alpha}\|x - (x_k + \nabla f(x_k))\|^2 \right) \\ &= \texttt{prox}_g(x_k + \alpha \nabla f(x_k)) \end{aligned}$$

Theorem 4.6 (Convergence rate of ISTA)**.** *Given $g : \mathbb{R}^n \to \mathbb{R}$ convex and continuous, $f : \mathbb{R}^n \to \mathbb{R}$ continuous differentiable with Lipschitz continuous derivative. Then there for any initial value x_0, the algorithm 4.4 converges to the minimum x_* in sublinear rate:*

$$F(x_k) - F(x_*) \leq \frac{L\|x_0 - x_*\|}{2k} \tag{II.74}$$

4.5 FISTA

5 The Coordinate Descent Method

5.1 The Basics

The problem formulation

$$\begin{aligned} &\min_{x \in \mathbb{R}^n} \hat{f}(x) + \lambda r(x) \\ &\texttt{s.t.} \\ &\hat{f}(x) = \frac{1}{M}\sum_{i=1}^{M} \phi_i(c_i^T x) \ \texttt{mean sum of convex scalar functions} \\ &r : \mathbb{R}^n \to \mathbb{R} \ \texttt{convex regularisation, e.g.,} \ r(x) = \sum_{i=1}^{n} r_i(x_i) \textbf{(separable case).} \\ &\lambda \geq 0 \ \texttt{regularisation parameter.} \end{aligned} \tag{II.75}$$

How to choose the i_k:

1. Fully random.
2. cyclic: $i_{k+1} = (i_k \mod n) + 1$.
3. Essentially Cyclic (within k iteration every component appear at least once).

Choice of the α_k:

1. ADMM
2. Method in nonlinear optimisation, such as backtracking.

In certain cases, the minimiser can be calculated explicitly:

1. **The proximal-point:**

$$z_{i_k} = \arg\min_{z\in\mathbb{R}} f(\underbrace{z}_{i_k\texttt{th component}}, x_{l,l\neq i_k}) + \lambda r_{i_k}(z) + \frac{1}{2\alpha_k}\|z - x_{i_k}\|^2. \tag{II.76}$$

2. **The linearised proximal point:** Linearisation around x_k we have

$$z_{i_k} = \arg\min_{z\in\mathbb{R}} f(x_k) + \langle \Big(\nabla f(x_k)\Big)_{i_k}, z - x_{i_k}\rangle + \lambda r_{i_k}(z) + \frac{1}{2\alpha_k}\|z - x_{i_k}\|^2 \tag{II.77}$$

Example (Linear curve fitting)**.** Given $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, observe the problem

$$\begin{aligned} &\min_{w\in\mathbb{R}^n} \frac{1}{2}\|w\|_2^2, \\ &Aw = b. \end{aligned} \tag{II.78}$$

Observe the Lagrange function:

$$L(w,\lambda) = \frac{1}{2}\|w\|_2^2 - \langle \lambda, Aw - b\rangle. \tag{II.79}$$

The necessary optimality condition gives

$$\nabla_w L(w,\lambda) = w - \langle \lambda, A\rangle \overset{!}{=} 0 \quad \Rightarrow \quad \langle A, \lambda\rangle = w.$$

Plug in $\langle A,\lambda\rangle = w$ in the Lagrange function:

$$L(A^T\lambda, \lambda) = -\frac{1}{2}\|A^T\lambda\|_2^2 + b^T\lambda$$

We now apply the duality theory, define w^* solution of

$$\begin{aligned} &\min_{w\in\mathbb{R}^n} \frac{1}{2}\|w\|^2, \\ &Aw = b. \end{aligned} \tag{II.80}$$

the the Lagrange function admits the minimiser at

$$\begin{aligned} &\max_{\lambda\in\mathbb{R}^m} L(A^T\lambda, \lambda) \\ &A^T\lambda = w^* \end{aligned}$$

which is equivalent to

$$\begin{aligned} &\min_{\lambda\in\mathbb{R}^m} \frac{1}{2}\|A^T\lambda\|^2 - b^T\lambda, \\ &A^T\lambda = w^*. \end{aligned} \tag{II.81}$$

Choose the step size $\alpha_k = 1$ and solve the minimisation problem w.r.t. λ we have the iteration

$$\lambda_{k+1} = \lambda_k - \Big(\langle A, A^T\lambda\rangle - b\Big)\cdot e_{i_k}.$$

with $w = A^T\lambda$ we have the iteration for w

$$w_{k+1} = w_k - A^T(\langle A, A^T\lambda\rangle - b)\cdot e_{i_k}.$$

Remark (Relation to the other algorithms). Relation to the other method:

1. **The SGD-method:**
2. **The SOR-method:**
3. **ADMM method**

The application

1. **The LASSO Problem:**
2. **SVM**

5.2 Technical Analysis of the CD-method.

Problem: Which class of functions admits convergence with CD-method?

Example (Powell: non-convergence for non-convex functions)**.**

Definition 5.1 (componentwise Lipschitz constant)**.** Given $f : \mathbb{R}^n \to \mathbb{R}$, f is componentwise Lipschitz differentiable if

$$\forall i \in \{1,\dots,n\} \exists L_i > 0 : |\nabla f(x+te_i)_i - (\nabla f(x))_i| \leq L_i|t| \quad \forall x \in \mathbb{R}^n, t \in \mathbb{R}. \tag{II.82}$$

we define accordingly

$$L_{max} := \max_{i\in\{1,\dots,n\}} L_i \tag{II.83}$$

L_{res} is called the **restricted Lipschitz constant** if

$$\|\nabla f(x+te_i) - \nabla f(x)\| \leq L_{res}|t| \quad \forall x \in \mathbb{R}^n, t \in \mathbb{R}. \tag{II.84}$$

Remark.

1. If f is Lipschitz differentiable, then we have

   $$L_{res} \leq L_{max} \leq L \leq nL_{max} \tag{II.85}$$

   We define accordingly

   $$\Lambda := \frac{L_{res}}{L_{max}} \tag{II.86}$$

2. If f is convex and twice differentiable, we have pointwise

   $$|(\nabla^2 f)_{ij}| \leq \sqrt{\nabla^2_{ii} f \cdot \nabla^2_{jj} f} \tag{II.87}$$

   therefore

   $$1 \leq \Lambda \leq \sqrt{n}. \tag{II.88}$$

3. If f is separable we have

   $$\Lambda = 1.$$

Assumption 5.1.
1. f is convex and Lipschitz differentiable with Lipschitz constant(of the derivative) L.
2. Given the initial value $x_0 \in \mathbb{R}^n$ and the set of global minimiser S, it satisfies

$$\begin{aligned} &\max_{x\in L_0} \min_{x_*\in S} \|x - x_*\| \leq R_0, \quad R_0 < \infty, \\ &\texttt{with}, \\ &L_0 := \{x \in \mathbb{R}^n | f(x) \leq f(x_0)\}. \end{aligned} \tag{II.89}$$

Assumption 5.2. The descent coordinate i_k is randomly chosen.

Theorem 5.1. *Under the Assumption 5.1 and 5.2 and the Algorithm 5.1 satisfies* $\alpha_k = \frac{1}{L_{max}}$. *Then the CD-method converges sublinear with*

$$\mathbb{E}f(x_k) - f_* \leq \frac{2nL_{max}R_0^2}{k} \tag{II.90}$$

*In the case where f is **also** strongly convex with the constant c we have the linear convergence:*

$$\mathbb{E}f(x_k) - f_* \leq \left(1 - \frac{c}{nL_{max}}\right)^k \left(f(x_0) - f_*\right). \tag{II.91}$$

Proof.

1. By the Lipschitz differentiability we have by (**??**)

$$\mathbb{E}_{i_k} f(x_{k+1}) - f(x_k) \leq -\frac{1}{2L_{max}} \mathbb{E}_{i_k} \|\nabla f(x_k)_{i_k}\|^2 = -\frac{1}{2nL_{max}} \sum_{i_k=1}^{n} \nabla f(x_k)_{i_k}^2 = -\frac{1}{2nL_{max}} \|\nabla f(x_k)\|^2.$$

Taking the expectation $\mathbb{E} := \mathbb{E}_{i_0,\ldots,i_k}$ we have

$$\begin{aligned}\mathbb{E}f(x_{k+1}) - f_* &\leq \mathbb{E}f(x_k) - f_* - \frac{1}{2nL_{max}} \mathbb{E}\|\nabla f(x_k)\|_2^2 \\ &\leq \mathbb{E}f(x_k) - f_* - \frac{1}{2nL_{max}} (\mathbb{E}\|\nabla f(x_k)\|)^2,\end{aligned}$$

hence the sequence $\mathbb{E}f(x_k) - f_*$ is monotone decreasing. Moreover by convexity we have

$$\begin{aligned}f(x_k) - f_* &\leq \langle \nabla f(x_k), x_k - x_* \rangle \leq \|\nabla f(x_k)\| \|x_k - x_*\| \\ &\leq \|\nabla f(x_k)\| R_0.\end{aligned}$$

2. The strongly convex case. Since f is Lipschitz differentiable we have

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2L_{max}} |\nabla f(x_k)_{i_k}|^2$$

taking the expectation and minus f_* we have

$$\mathbb{E}f(x_{k+1}) - f_* \leq \mathbb{E}f(x_k) - f_* - \frac{1}{2nL_{max}} \|\nabla f(x_k)\|^2$$

By strongly convexity we have

$$\|\nabla f(x_k)\|^2 \geq 2c(f(x_k) - f(x_*)).$$

Plug in the last equation we have

$$\begin{aligned}\mathbb{E}f(x_{k+1}) - f_* &\leq (\mathbb{E}f(x_k) - f_*) - \frac{c}{nL_{max}} (\mathbb{E}f(x_k) - f_*) \\ \mathbb{E}f(x_{k+1}) - f_* &\leq (1 - \frac{c}{nL_{max}})(\mathbb{E}f(x_k) - f_*)\end{aligned}$$

□

Theorem 5.2. *Under the Assumption 5.1 and Assumption 5.2, the algorithm 5.1 converges sublinearly with*

$$\mathbb{E}f(x_k) - f_* \leq \frac{n(L_{max} + L)R_0^2}{2k}. \tag{II.92}$$

Proof. □

Theorem 5.3. *Under the assumption 5.1 and choose $\alpha_k = \frac{1}{L_{max}}$, choose the descent coordinate by*

$$\begin{aligned} i_0 &= 0, \\ i_{k+1} &= (i_k \mod n) + 1. \end{aligned} \tag{II.93}$$

Then the algorithm 5.1 converges sublinearly with

$$f(x_k) - f_* \leq \frac{4n(n+1)LR_0^2}{k+8} \tag{II.94}$$

If f is additionally strongly convex, then we have

$$f(x_k) - f_* \leq \left(1 - \frac{c}{2L_{max}(1+\frac{L^2}{L_{max}^2})}\right)^{\frac{k}{n}} (f(x_0) - f_*) \tag{II.95}$$

Proof. Beck: on the convergence of block coordinate descent methods. □

5.3 Extension to regularised case

Now we observe the problem:

$$\begin{aligned} &\min_{x\in\mathbb{R}^n} f(x) + \lambda r(x) := h(x), \\ &\texttt{with} \\ &\lambda > 0, \quad r = \sum_{j=1}^{n} r_i(x_i) \ \texttt{separable}. \end{aligned} \tag{II.96}$$

Assumption 5.3. h satisfies:

1. $f : \mathbb{R}^n \to \mathbb{R}$ Lipschitz differentiable with the Lipschitz constant L.
2. f is strongly convex with the constant c.
3. r_i is convex for all i.
4. h admits the minimiser on x_*.

Theorem 5.4. *Under the assumption 5.2 and assumption 5.3, the algorithm 5.1 converges* ***r-linearly*** *in expectation with*

$$\mathbb{E}h(x_k) - h_* \leq \left(1 - \frac{c}{nL_{max}}\right)^k \left(h(x_0) - h_*\right). \tag{II.97}$$

Remark. Compared with the unregularised case, the Lipschitz differentiability of h is not assumed.

Proof. Define the minimiser of the linear search

$$z_{k,i} := \arg\min_{z\in\mathbb{R}} \langle \nabla f(x_k)_i, z - (x_k)_i \rangle + \frac{1}{2} L_{max}(z - (x_k)_i)^2 + \lambda r_i(z).$$

□

Remark.

1. The strong convexity of f can be relaxed to the convexity. The convergence rate stays the same.
2. For non-convex f we can show the **sublinear** convergence.

5.4 Comparison of ansatzs

Example.

$$
\begin{aligned}
&f(x) = \frac{1}{2}x^T Qx \\
&\texttt{with} \\
&Q \ \texttt{SPD}
\end{aligned}
\tag{II.98}
$$

We construct Q as such

$$
\begin{aligned}
&Q = V_{r,\eta}\Sigma V_{r,\eta}^T + \xi I_n I_n^T, \\
&\Sigma \ \texttt{diagonal}, \\
&V_{r,\eta} = \eta V + (1-\eta)E_r
\end{aligned}
$$

with V random and orthogonal,

Setting the problems

Method of choosing the descent direction

- **IID:** choose and put back
- **EPOCH:** choose and not put back
- **CYCLIC:** cyclical(deterministic)

Method of choosing the step seize

- $\alpha_k = \frac{1}{L_{max}}$.
- α_k exact(f quadratic and convex).

Results

1.

6 The Clustering And Classification

$$d(x,y) = \|x-y\|_2^2, \nabla d(x,y) = \begin{bmatrix} 2(x-y) \\ 2(y-x) \end{bmatrix}$$

$$w_{ij} = \begin{cases} 1 & x_j \in C_i \\ 0 & \texttt{else} \end{cases}$$

$$\sum_{i=1}^{k} \sum_{x \in C_i} d(x_i, m_i) = \sum_{i=1}^{k} \sum_{j=1}^{N} d(x_j, x_i)$$

with that we obtain the following optimisation problem:

$$\min_{w \in \mathbb{R}^m}$$

Lemma 6.1. *Let $f : \mathbb{R}^n \to \mathbb{R}, f(x) = \frac{1}{2}x^T Ax + b^T x + c$, with A SPD, then we have*

$$\texttt{prox}_f(y) = \arg\min_x (f(x) + \|x-y\|_2^2 = (A+I)^{-1}(y-b)$$

Proof. □

6.1 SVM

Definition 6.1 (margin)**.** Given a sample $\mathcal{D} := \mathcal{S}(\omega) \in \mathbb{R}^d$ and a hard classifier $h(x) = \text{sign}(\langle w, x \rangle + b)$,

Lemma 6.2.

$$\text{dist}($$

Lemma 6.3. *Given the sample $\mathcal{D} \in (\mathbb{R}^d \times \{\pm 1\})^n$,*

Definition 6.2 (support vector)**.**

Theorem 6.4 (the quadratic optimization)**.**

Corollary 6.4.1.

6.2 The Kernel Method

Definition 6.3 (kernel function)**.** Let $X \neq \emptyset$ be a set, a function $k : X \times X \to \mathbb{R}$ is called a kernel function if there exits a Hilbert space H and a map $\varphi : X \to H$, such that

$$k(x,y) = \langle \varphi(x), \varphi(y) \rangle_H$$

Remark.

Given $A \in \mathbb{R}^{p \times n}, b \in \mathbb{R}^n$.

$x_0 \in \mathbb{R}^n$, $\lambda > 0$. $k = 0, 1, \ldots$ choose α_k. $x_{k+1} = \mathcal{S}_{\alpha_k \lambda}(x_k - 2\alpha_k A^T(Ax - b))$.

Given the function f, r, λ as above.

$x_0 \in \mathbb{R}^n$, $i_0 \in \{1, \ldots, n\}$ $k = 0, 1, \ldots$ choose i_k choose α_k $x_{k+1} = x_k - \alpha_k \nabla f(x_k)_{i_k}$

Given $f : \mathbb{R}^n \to \mathbb{R}$ as before(mean sum of convex functions), r separable, $\lambda > 0$.

$x_0 \in \mathbb{R}^n$ $k \leq n$ choose $i_k \in \{1, \ldots, n\}$. calculate z_{i_k} proximal-point or linearised proximal. $x_{k+1} = x_k + (z_{i_k} - (x_k)_{i_k})e_{i_k}$

Chapter III

Markov Process and Reinforcement Learning

1 Markov Decision Process

discrete time stochastic optimization problem.

$$V^i(s,\Pi^i,\Pi^{-i}) := \max_a \mathbb{E}_s \sum_{t=0}^{\infty} \gamma^t r(s_t^i, a_t^i, (s_t^{-i}), (a_t^{-i}))$$

where s the starting point, s_t denote the state and a_t the action at time t, $\gamma \in (0,1)$ the discounting factor. a_t^{-i}, s_t^{-i} the collective of the extended players. $\hat{\Pi}$ is the Nash equilibrium, if

$$\forall i = 1,\dots,N : V^i(s,\hat{\Pi}^i,\hat{\Pi}^{-i}) \geq V^i(s,\Pi^i,\Pi^{-i}) \quad \forall \Pi$$

$$V^*(s,) = r(s,a) + \sum_{s'} p_s(s'|s,a) \cdot V^*(s').$$

assumptions

1. finite state
2. $V^n \to V^*$,
3. policy iteration $\Pi^n \to V^{\Pi_n} \to \Pi^{(n+1)}$.

Now: reinforcement learning.

```
input:a
output:states', r(a,s)
```

Intuitively, for given policy Π, $\frac{1}{N}\sum_1^N \delta_{S_t^j} \to \mu_t$ where μ_t is a deterministic measure flow.
propagation of Chaos.

1.1 Formulations

Definition 1.1 (Markov decision process)**.** Given the state space S, A the set of actions, $p_s(ds|s,a) : S \times A \to S$ the transition kernel for $s \in S$ and $a \in A$. For simplicity we assume the state space and action space to be countable. Moreover let $\bar{r}(s,a) := \int_{\mathbb{R}} r' p_r(dr'|s,a)$ be an $\mathbb{R}$-valued reward function given the reward transition kernel $p_r(dr|s,a)$, where $p_r(dr'|s,a) = \delta_{\bar{r}(s,a)}(dr')$.

Assumption 1.1.

1. discrete time
2. infinite time horizon

Definition 1.2 (finite Markov decision process)**.** A Markov decision process is called `finite` if S and A are finite.

Definition 1.3 (policy(strategy))**.**

1. A policy $\pi = (\pi_t)$ is given by $\pi_t : S \to P(A)$
2. (π_t) is deterministic if

$$\forall s \in S \forall t \geq 0 \exists a' \in A : \quad \pi_t(s) = \delta_{a'}(da),$$

   in this case we identify $\pi_t(s) = a'$ with map $S \to A$.
3. π is stationary if $\pi_t = \pi_u, \forall t, u \geq 0$.

The optimization problem under consideration is the single agent problem

$$\max_{\pi} \mathbb{E}_s \left(\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t = \pi_t(s_t)) \right) \tag{III.1}$$

with $\gamma \in [0, 1)$.

Remark.

1. We assume the reward function to be bounded
2. In finite MDP, we can w.l.o.g. consider the deterministic reward.

Definition 1.4 (policy value)**.** The optimal value function V is defined as

$$V_\Pi(s) = \mathbb{E}_\pi[\sum \gamma^t r(s_t, a_t) | s_0 = s]$$

Remark. Alternative optimal objective

1. some subset $S^+ \subset S$ of cemetery state where the game terminates(at stopping time).
2. ergodic control ("optimised time-average"):

$$\frac{1}{T} \mathbb{E}[\sum_{t=0}^{T} r(s_t, a_t)]$$

   or

$$\lim_{\gamma \uparrow 1} \frac{1-\gamma}{\gamma} \mathbb{E}[\sum^{\infty} \gamma^t r(s_t, a_t)]$$

Remark (precise stochastic modelling). On the canonical probability space which is $(S \times A)^{\mathbb{N}_0}$ for deterministic reward and $(S \times A \times R)^{\mathbb{N}_0}$ for the random rewards. (Lonescu-Talcea).

1.2 Optimal policy for MDP

Assumption 1.2. infinite time horizon, stationary policies, bounded reward.

Under the assumption we have

$$V^*(s) := \sup_{\pi} V_\pi(s) < C < \infty$$

Definition 1.5 (the optimal policy)**.** The policy π^* is optimal if

$$\forall \pi \forall s: \quad V^*(s) \geq V_\pi(s)$$

Definition 1.6 (the Q-value function)**.** The state-action value function Q_π for policy π

$$\begin{aligned} Q_\pi(s,a) &:= \mathbb{E}_{s,a,\pi}[r(s,a)] + \mathbb{E}_{s,a,\pi}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t,a_t) | s_0 = s, a_0 = a\right] \\ &= \bar{r}(s,a) + \mathbb{E}_{s,a,\pi}[\gamma V_\pi(s_1)] \end{aligned}$$

Theorem 1.1. *For Π, Π', if*

$$\forall s \in S: \quad \mathbb{E}_{s,\Pi'}[Q_\Pi(s,a_0)] \geq \mathbb{E}_{s,\Pi}[Q_\Pi(s,a_0)],$$

then it holds

$$\forall s \in S: \quad V_{\Pi'}(s) \geq V_\Pi(s)$$

Proof. □

Remark.

1. The result holds for finite horizon Markov processes.
2. A policy π is optimal if and only if

$$\forall s \in S \quad a \in A: \quad a \in \arg\max_{a'} Q_\pi(s,a') \texttt{ with positive probability, i.e., } \pi(s)(a) > 0.$$

Theorem 1.2. *If π is a optimal policy, then it holds*

$$\forall (s,a) \in S \times A \ \ \texttt{with} \ \ a \in \arg\max_{a'} Q_\pi(s,a'): \quad \pi(s)(a) > 0.$$

Proof. Argue by contradiction. □

Theorem 1.3. *1. Any finite MDP, i.e., $|S| < \infty$ admits an optimal policy, which is deterministic.*

2.

Proof. The finite state implies

$$\sup_\pi \sum_s V_\pi(s) < \infty.$$

Since the number of deterministic policies are also finite, it holds over *pi* determinsitic

$$\exists \pi^*: \quad \pi^* = \arg\max_\pi \sum_s V_\pi(s)$$

claim: π^* also optimal over random policy, indeed, let π, π' be some randomised policies, it holds

$$\mathbb{E}_{s,\pi'}[Q_\pi(s,a_0)] = \int_A \pi'(s)(da_0) \cdot \left(r(s,a_0) + \int_S p_s(ds_1|s,a_0)\gamma V_\pi(s_1)\right).$$

For policy π, let $^h at \pi : S \to A$ deterministic as:

$$\hat{\pi}(s) \in \arg\max_a Q_\pi(s,a)$$

then

$$Q_\pi(s, a_0 \sim \hat{\pi}(s)) \geq \max_a Q_\pi(s,a) \geq Q_\pi(s, a_0 \sim \pi(s)) \quad a.e. \forall s \in S$$

hence

$$\mathbb{E}_{s,\hat{\pi}}[Q(s,a_0)] \geq \mathbb{E}_{s,\pi}[Q(s,a_0)]$$

□

Assumption 1.3. Finite MDP and deterministic policy $\pi : S \to A$.

Remark.

1. Let π^* be the optimal policy and denote $Q^* := Q_{\pi^*}$, $V^* := V_{\pi^*}$.
2. By Bellman we know $\pi^* \in \arg\max_a Q^*(s,a)$ for all $s \in S$.
3. Indeed, by the argument of the last theorem we can always choose a deterministic policy with $\pi(s) \in \arg\max_a Q^*(s,a)$, such that π is optimal.
4. Hamiltonian-Jacobi-Bellman of dynamic programming

$$V^*(s) = \mathbb{E}_{s,\pi^*}\left[Q^*(s,a_0)\right] = r(s,\pi^*(s)) + \gamma \mathbb{E}_{s,\pi^*}\left[V^*(s_1)\right] = \max_a \{r(s,a) + \gamma \sum_{s' \in S} p_s(s'|s,a)V^*(s')\}.$$

1.3 Policy Evaluation

Theorem 1.4. *For a fixed policy π the value function is a linear system of equations of V.*

Remark. Can identify function $S \to \mathbb{R}$ with vector $\mathbb{R}^S$.

$$V = R + \gamma PV \texttt{ with } R = r(s,\pi(s))_{s \in S}$$

Theorem 1.5. *For finite MDP the linear system admits a unique solution of V.*

Proof. P is a stochastic matrix, hence $(I - \gamma P)$ invertible for $\gamma \in (0,1)$. Solve the linear system for V:

$$(I - \gamma P)V = R.$$

□

Theorem 1.6. *The value iteration map is a contraction with*

$$\|\phi(U) - \phi(V)\|_\infty \leq \gamma \|U - V\|_\infty \quad \gamma \in (0,1).$$

Proof. □

Remark.

$$\phi V_\pi(s) = \max_{a \in A} Q_\pi(s,a).$$

Theorem 1.7 (Bellman optimality principle)**.**

2 Functional space and transformation

The infinite-width shallow neurone network $f(x) = \sum_{i=1}^\infty c_i \sigma(a \cdot x - b)$ can be regarded as a measure on the domain $\Omega = \mathbb{S} \times \mathbb{R}$, constructed by countable many atoms. We denote this measure by μ and the space finite measure on Ω as $M(\Omega)$. Observe $f \in C(\Omega)$ belongs to de dual space of $M(\Omega)$. Indeed, it relates closely to the dual of μ by the Radon transformation and Laplacian.

3 The Radon Transform

Definition 3.1 (the Radon transform)**.** Given $f \in L^1(\mathbb{R}^d)$, for $a \in \mathbb{R}^d, b \in \mathbb{R}$, define the hyperplane $X(a,b) := \{x \in \mathbb{R}^d | a \cdot x + b = 0\}$ define the mapping

$$\mathcal{R}[f] : \mathbb{R}^{d+1} \to \mathbb{R}$$
$$\mathcal{R}[f](a,b) := \int_{x \in X(a,b)} f(x) \nu_{d-1}, \quad \forall (a,b) \in \mathbb{S}^d \times \mathbb{R}.$$

Remark. We consider in the following $f(x) = (a \cdot x - b)$ with σ the ReLU activation function. Observe ∇f is the Heaviside function and Δf does the point landing at (a,b).

Definition 3.2 (representation cost)**.** Given the (infinite-width) two-layer ReLU net $g_\theta : \mathbb{R}^d \to \mathbb{R}$ parametrised by θ:

$$g_\theta(x) := \sum_{i=1}^{k} c_i \sigma(a_i \cdot x_i + b_i) + d,$$

`with`

$$\theta := \{A := [a_1, \ldots, a_k] \in \mathbb{R}^{k \times d}, b := [b_1, \ldots, b_k] \in \mathbb{R}^k, c := [c_1, \ldots, c_k] \in \mathbb{R}^k, d \in \mathbb{R} : | \quad k \in \mathbb{N}\}.$$

For any $\theta \in \Theta$, we define the Euclidean weights norm that excludes the bias term b, d:

$$C(\theta) := \frac{1}{2}\left(\|A\|_F^2 + \|c\|^2 \right).$$

and consider the minimal representation cost to the function $f : \mathbb{R}^d \to \mathbb{R}$ that is **exactly** representable by the neural network:

$$R(f) := \inf_{\theta \in \Theta} C(\theta) \ \ \texttt{s.t.} \quad f = g_\theta.$$

By the homogeneity of the ReLU function, the infimum is attained at

$$\Theta' := \{\theta \in \Theta : \quad \|A\|_F^2 = 1\}$$

$$\bar{R}(f) := \lim_{\varepsilon \to 0} \tag{III.2}$$

Definition 3.3 (the Hilbert transformation)**.** Give $g : \mathbb{R} \to \mathbb{R}$, define

$$\mathcal{H}[g](b) := \frac{1}{\pi} \textbf{p.v.} \int_{-\infty}^{\infty} \frac{g(z)}{b - z} dz.$$

Lemma 3.1. *Suppose $f = h_{\alpha,v,c}$ for some $\alpha \in \mathcal{S}(\mathbb{S} \times \mathbb{R})$ with α even measure, $v \in \mathbb{R}^d$, $c \in \mathbb{R}$. Then the measure α can be recovered by*

$$\alpha = -\gamma_d \mathcal{R}[(-\Delta)^{(d+1)/2} f]$$

and the representation cost of f is

$$\bar{R}_1(f) = -\gamma_d \|\mathcal{R}[(-\Delta)^{(d+1)/2} f\|_1.$$

4 Reinforced Learning

Motivation:

5 mean field games

5.1 finite population games

Mean field games can be perceived as the large population limits $N \to \infty$.

Assumption 5.1.

1. discrete time,

2. finite time horizon T, $t = 0, \ldots, T$,

3. finite state and action sets $\mathcal{S}$ and $\mathcal{A}$,

4. for finite set $\mathcal{X}$, with $\mathcal{B}(2^{\mathcal{X}})$, let $\mathcal{P}$ be the probability measure on $\mathcal{X}$,

5. transition dynamics, for each agent, described by transition kernel $p_s = p$ and the initial distribution $\mu_0 \in \mathcal{P}(\mathcal{S})$.

For $|\mathcal{X}| = 3$ it holds $\mathcal{P}(\mathcal{X}) = \Delta^{\mathcal{X}}$.